ME751

Advanced Computational Multibody Dynamics

[Spring 2019]

Lecture 24

10/30/2019

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The quote the day

“Son, if you really want something in this life, you have to work for it. Now quiet! They're about to announce the lottery numbers.”
Homer Simpson

Introduction to Computational Fluid Dynamics

1. Review and Background
   - Notations, Gradient, Divergence, Eulerian and Lagrangian Description
2. Conservation Principles
   - Conservation law, advective and diffusive fluxes, advection-diffusion equation
3. Space Discretization
   - Weighted Residual method, finite-difference, finite-volume, [finite-element], and SPH
4. Fluid Flows
   - Mass and Momentum conservation, time-integration, explicit vs. implicit
5. Fluid-Solid Interaction

Assumptions

1. Matters are composed of discrete objects such as molecules, particles, etc.

2. Studying every system via a discrete point of view is not possible, e.g., fluids

3. We are assuming that the system of interest is a **continuum**

4. In continuous systems mass is distributed evenly, contrary to the discrete systems that have been studied in the course so far

5. Sometimes discrete systems may be approximated via continua, e.g., granular material

Notations

For any point in a continuum, we may define variables that are different in nature:

1. Scalars (zero-order tensor):
   - 3^0=1 elements, pure numbers, imply only a sense of magnitude, e.g., temperature, density, etc.
   - They may be dimensional, e.g., $T = 55°F$
   - **Notation**: non-bold lower-case or non-bold upper-case, e.g. T, ρ, m

2. Vectors (first-order tensor):
   - 3^1=3 elements in 3D, imply a sense magnitude and direction, e.g., position, velocity, and acceleration
   - May have dimensions, e.g., $\boldsymbol{u} = [1.0, 0.2, 0.5]^T m/s$
   - **Notation**: bold and lower-case Latin letters, e.g. $\boldsymbol{x}, \boldsymbol{u}$

3. Tensors (second-order tensor):
   - 3^2=9 elements in 3D, imply a sense of direction and magnitude depending on a direction
   - They may be dimensional, e.g., $\boldsymbol{\sigma} = \begin{bmatrix} 1 & 0.2 & 0.3 \\ 0.2 & 0.5 & 0 \\ 0.3 & 0 & 0.2 \end{bmatrix} N/m^2$
   - **Notation**: bold and upper-case Latin letters and bold and lower-case Greek e.g. $\mathbf{S}, \mathbf{E}, \boldsymbol{\sigma}, \boldsymbol{\tau}$

Background – Gradient Operator

- Gradient implies a sense of change in different directions
- For instance, the temperature gradient is a measure for how much the temperature changes if you move in different directions
- The gradient of a field is the change of that field with respect to change in space
- For instance, $\frac{\partial T}{\partial x}$, $\frac{\partial T}{\partial y}$ are the notation we use to measure the rate of change of T in the x and y directions, respectively

Background – Gradient Operator

- Mathematically, we define the gradient of a **scalar** such as the temperature to be a **vector** :

$$\nabla T = (\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z})$$

- Mathematically, we define the gradient operator as:

$$\nabla = (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})$$

- If you are interested in the gradient in any direction $\mathbf{n} = \left(n_x, n_y, n_z\right)^T$, you can multiply (dot product) ∇T with the unit vector $\boldsymbol{n}$

$$\nabla T\Big|_{\boldsymbol{n}} = \nabla T \cdot \boldsymbol{n} = \frac{\partial T}{\partial x} n_x + \frac{\partial T}{\partial y} n_y + \frac{\partial T}{\partial z} n_z$$

Background – Gradient Operator

- If you are interested in the gradient in any direction $\mathbf{n} = \left(n_x, n_y, n_z\right)^T$, you can multiply (dot product) ∇T with the unit vector $\boldsymbol{n}$

$$\nabla T\Big|_{\boldsymbol{n}} = \nabla T \cdot \boldsymbol{n} = \frac{\partial T}{\partial x} n_x + \frac{\partial T}{\partial y} n_y + \frac{\partial T}{\partial z} n_z$$

- For instance, with $\boldsymbol{n} = (1,0,0)^T$ we expect to recover the $\frac{\partial T}{\partial x}$ from the gradient

- Note that the gradient of a scalar (zero-order tensor) is a vector (first-order tensor)

Background – Gradient Operator

- We could think about the gradient of a vector such as $\boldsymbol{u} = \left(u_x, u_y, u_z\right)^T$ in a similar fashion
- The gradient of a vector field (first-order tensor) is a tensor field (second-order tensor)
- Think of gradient of a vector as a gradient of each component of a vector

$$\nabla \boldsymbol{u} = \nabla \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} & \frac{\partial u_x}{\partial z} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} & \frac{\partial u_y}{\partial z} \\ \frac{\partial u_z}{\partial x} & \frac{\partial u_z}{\partial y} & \frac{\partial u_z}{\partial z} \end{bmatrix}$$

Background – Gradient Operator

- Again, if the gradient in a particular direction is required you can multiply by the unit vector in that direction
- For instance,

$$\nabla \boldsymbol{u}\Big|_{\boldsymbol{n}} = \nabla \boldsymbol{u} \cdot \boldsymbol{n} = \begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} & \frac{\partial u_x}{\partial z} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} & \frac{\partial u_y}{\partial z} \\ \frac{\partial u_z}{\partial x} & \frac{\partial u_z}{\partial y} & \frac{\partial u_z}{\partial z} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial u_x}{\partial x} n_x + \frac{\partial u_x}{\partial y} n_y + \frac{\partial u_x}{\partial z} n_z \\ \frac{\partial u_y}{\partial x} n_x + \frac{\partial u_y}{\partial y} n_y + \frac{\partial u_y}{\partial z} n_z \\ \frac{\partial u_z}{\partial x} n_x + \frac{\partial u_z}{\partial y} n_y + \frac{\partial u_z}{\partial z} n_z \end{bmatrix}$$

Background – Divergence Operator

- Divergence operator provides information about **sinks** or **sources**

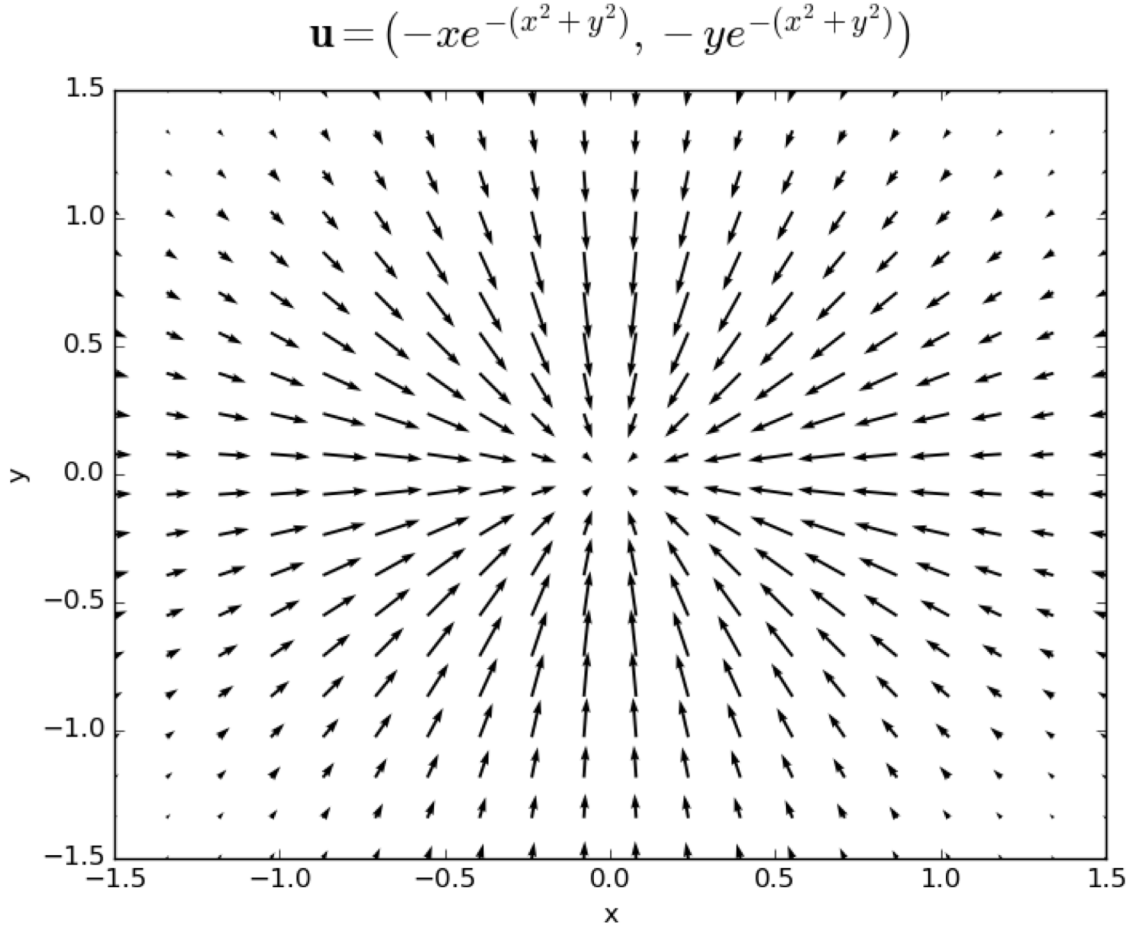
- If divergence operator is applied on a **vector** such as $\boldsymbol{u} = \left(u_x, u_y, u_z\right)^T$ the results is a **scalar**

$$\nabla \cdot \boldsymbol{u} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot \left(u_x, u_y, u_z\right)$$
$$= \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}$$

- Positive divergence indicates the presence of a **source (**expansion in the field)
- Negative divergence suggests the presence of a **sink (**contraction in the field)
- Zero divergence : no sink or source

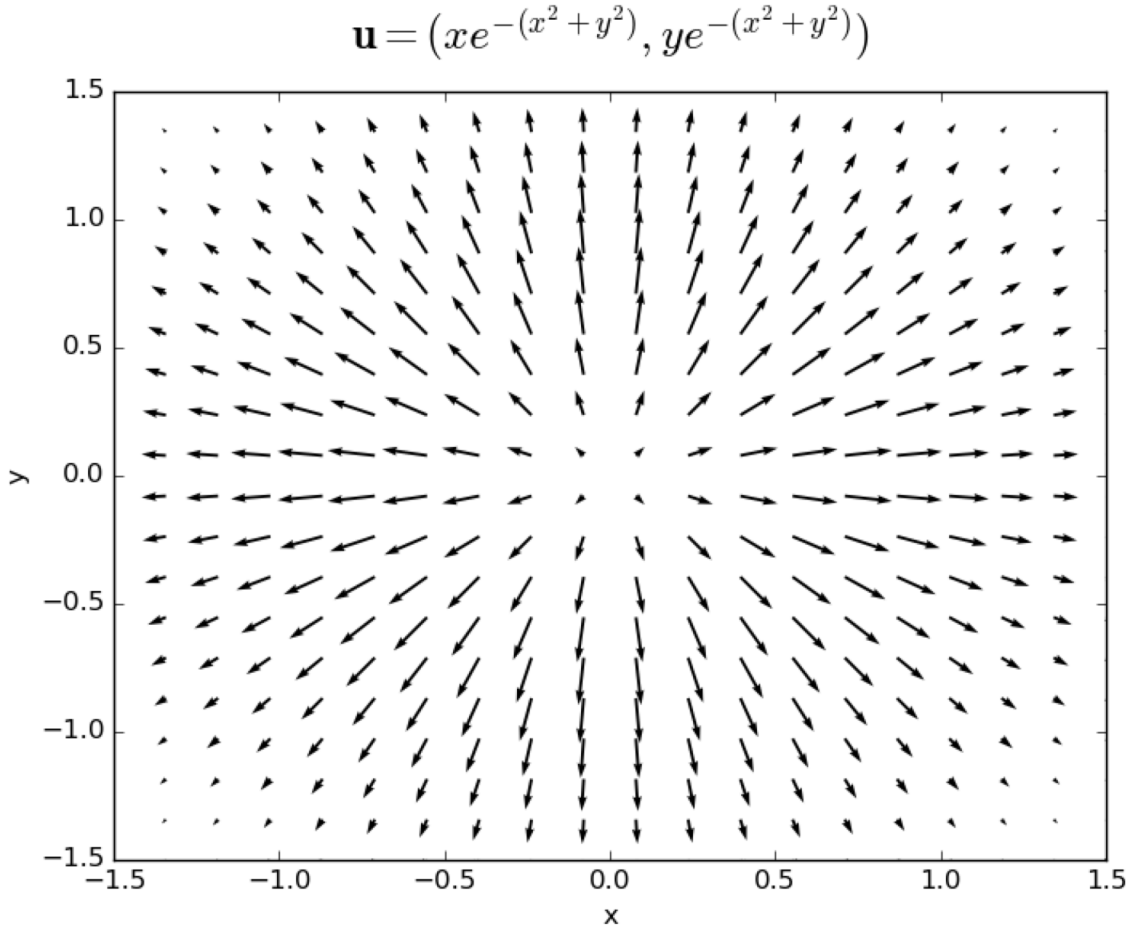
Background – Divergence Operator

- $\nabla \cdot \boldsymbol{u}|_{(0,0)} = \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y}\right)|_{(0,0)} = -2$ is negative, as if there is a sink in the origin



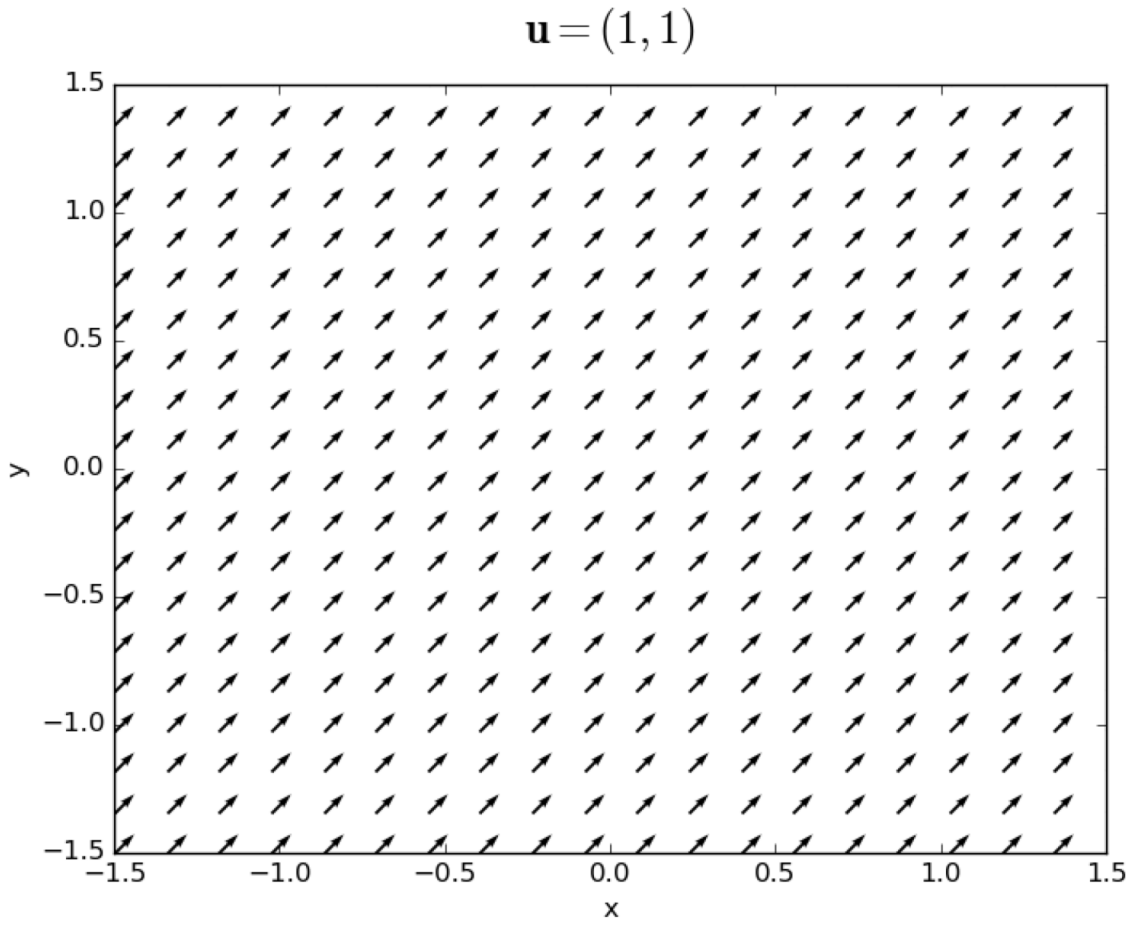
Background – Divergence Operator

- $\nabla \cdot \boldsymbol{u}|_{(0,0)} = \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y}\right)|_{(0,0)} = +2$ is positive, as if there is a source in the origin



Background – Divergence Operator

- $\nabla \cdot \boldsymbol{u}|_{(0,0)} = \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y}\right)|_{(0,0)} = +0$, no source/sink in the origin



Background – Divergence Operator

- Divergence operator provides information about **sinks** or **sources**

- If divergence operator is applied on a **tensor,** the results is a **vector**

$$\nabla \cdot \boldsymbol{S} = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}] \cdot \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} = \begin{bmatrix} \frac{\partial S_{11}}{\partial x} + \frac{\partial S_{12}}{\partial y} + \frac{\partial S_{13}}{\partial z} \\ \frac{\partial S_{21}}{\partial x} + \frac{\partial S_{22}}{\partial y} + \frac{\partial S_{23}}{\partial z} \\ \frac{\partial S_{31}}{\partial x} + \frac{\partial S_{32}}{\partial y} + \frac{\partial S_{33}}{\partial z} \end{bmatrix}$$

- You may make the same argument about the sign of each component of the vector $\nabla \cdot \boldsymbol{S}$, and say if there exists a **vector** sink/source in the domain

Eulerian and Lagrangian Descriptions

- In analyzing the behavior of continua you may think of two types of degrees of freedom
  - The degree of freedom is stationary, and it is "sensing" what is happening where it is located (**Eulerian)**
  - The degree of freedom is moving, potentially with the flow velocity (**Lagrangian**)

- We are interested in calculating the time-derivative of functions of form $f = f(\boldsymbol{x}, t)$
  - Note that $\boldsymbol{x}$ is a vector representing (x, y, z)
  - Eulerian time derivative: If the position of the observer is time-independent
  - Lagrangian time derivative: If the position of the observer is time-dependent, i.e. $\boldsymbol{x} = \boldsymbol{x}(t) = (x(t), y(t), z(t))$

Eulerian and Lagrangian Descriptions

- Remember the derivative of Multivariate function $f = f(\boldsymbol{x}(t), t) = f(x(t), y(t), z(t), t)$, and the chain rule:

$$\begin{aligned}
\frac{Df}{Dt} &= \frac{df(x(t), y(t), z(t), t)}{dt} \\
&= \left(\frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz\right)/dt \\
&= \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt} \\
&= \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}u_x + \frac{\partial f}{\partial y}u_y + \frac{\partial f}{\partial z}u_z \\
\frac{Df}{Dt} &= \frac{\partial f}{\partial t} + \boldsymbol{u} \cdot \nabla f
\end{aligned}$$

- Above, $\boldsymbol{u} = (u_x, u_y, u_z) = (\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt})$

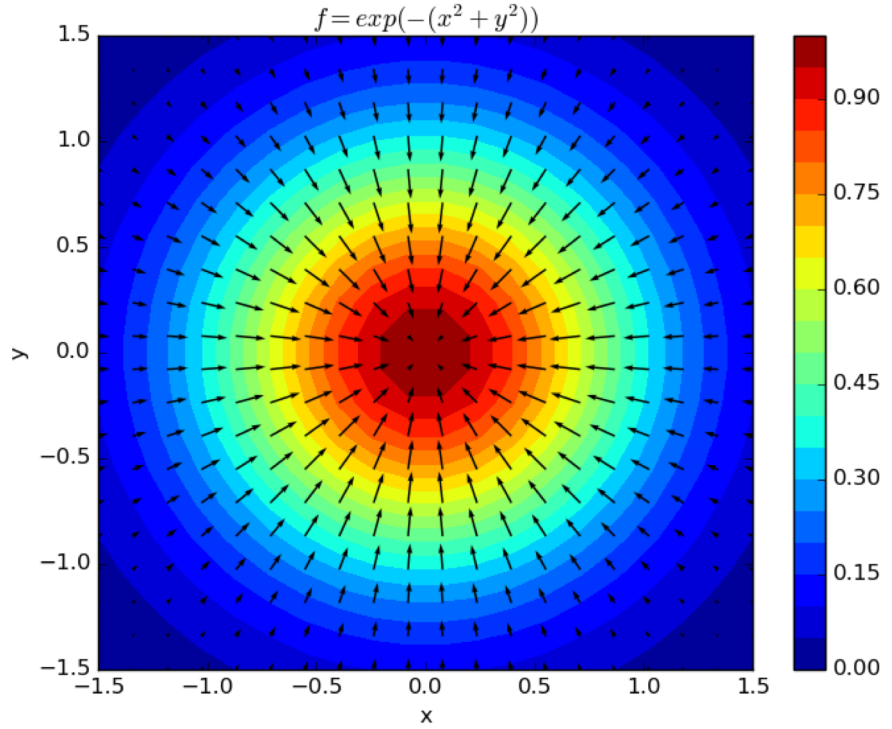
Eulerian and Lagrangian Descriptions

$$\underbrace{\frac{Df}{Dt}}_{\text{Material/Lagrangian derivetive}} = \underbrace{\frac{\partial f}{\partial t}}_{\text{Eulerian derivative}} + \underbrace{\boldsymbol{u} \cdot \nabla f}_{\text{Advective derivative}}$$

- If the observer is not moving, i.e., $\boldsymbol{u} = \mathbf{0}$, then the time-derivative is the same in both descriptions
- What is the physical meaning of this equation?

Eulerian and Lagrangian Descriptions

- Imagine a source (heat, concentration, etc.) in the field $f = \exp(-(x^2 + y^2))$
- The function value is illustrated with color, blue (low) to red (high)
- The function gradient is illustrated with a vector field in this figure
- Imagine an observer located at $\boldsymbol{x} = (1,0)$
- The gradient of f at the location of the observer is $\nabla f|_{(1,0)} = (-0.735,0)$

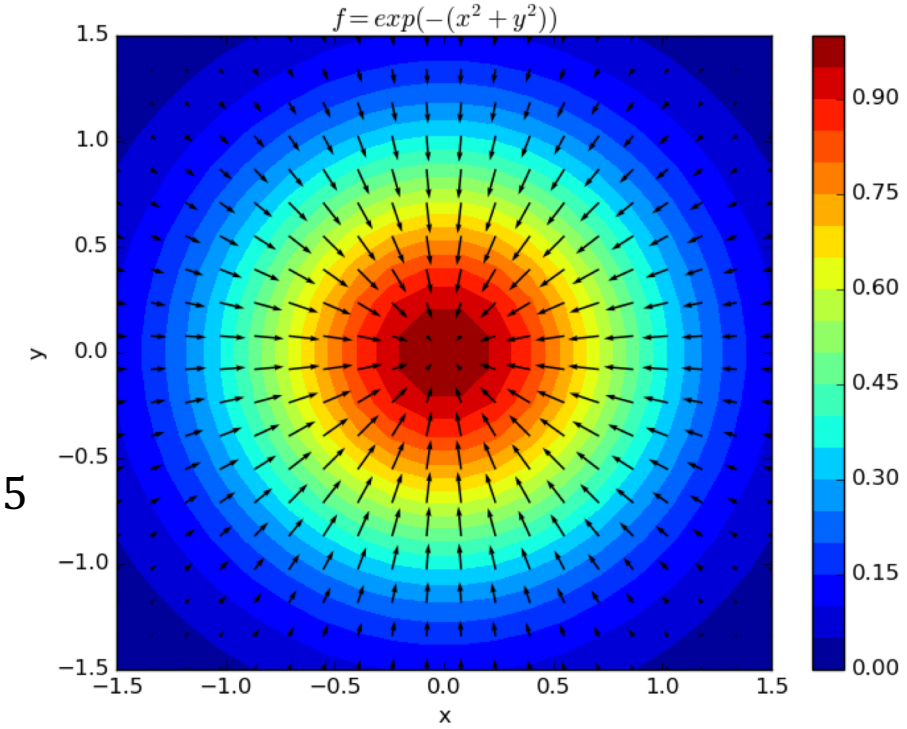


Eulerian and Lagrangian Descriptions

- If the observer is stationary, it does not sense any change over time, because $f \neq f(t)$

$$\underbrace{\frac{Df}{Dt}}_{\text{Material/Lagrangian derivetive}} = \underbrace{\frac{\partial f}{\partial t}}_{\text{Eulerian derivative}} + \underbrace{\boldsymbol{u} \cdot \nabla f}_{\text{Advective derivative}}$$

- If the observer is moving, you would expect the observer to sense some change :
  - If it is moving toward the source, a positive change over time
  - If it is moving away from the source, a negative change over time
  - This advective rate of change is manifested in the $\boldsymbol{u} \cdot \nabla f$ term
  - E.g., $\boldsymbol{u} = (-1,0)$, i.e., moving toward the source, $\frac{Df}{Dt} = +0.735$
  - E.g., $\boldsymbol{u} = (+1,0)$, i.e., moving away from the source, $\frac{Df}{Dt} = -0.735$
  - E.g., $\boldsymbol{u} = (0,+1)$ or $\boldsymbol{u} = (0,-1)$, the observer is not really moving toward the source or away from it because it is moving perpendicular to the direction of the gradient, i.e., $\boldsymbol{u} \cdot \nabla f = 0$ in both cases. Hence, $\frac{Df}{Dt} = 0.0$



Eulerian and Lagrangian Descriptions

- Note that there is a distinction between Eulerian/Lagrangian, and being meshless/mesh-based
  - A mesh-based method can be used in a Lagrangian fashion or Eulerian fashion, e.g., Finite Element is used in its Lagrangian form in structural mechanics, while it follows the Eulerian description in CFD
  - A meshless discretization can be used in an Eulerian framework

- Often, however, the word mesh-less discretization is associated with the Lagrangian description and mesh-based discretization is associated with Eulerian discerption

Conservation Principles

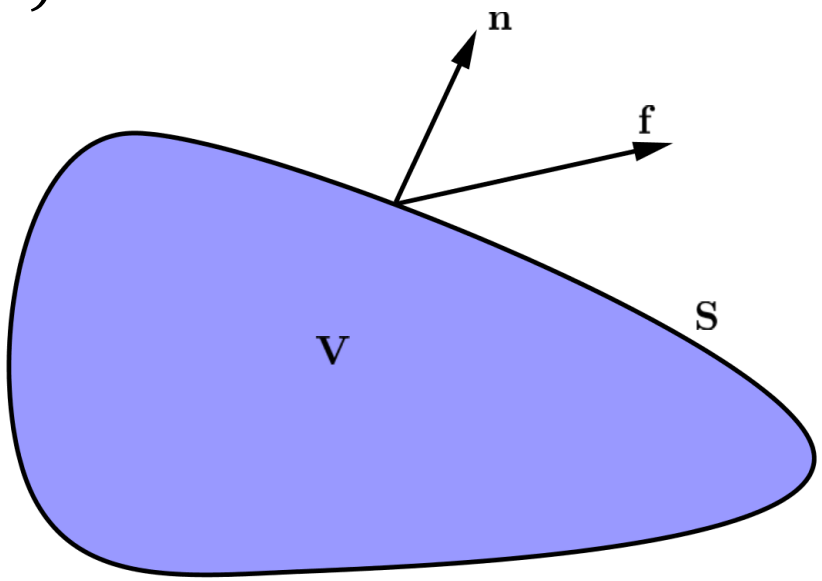
- The important question we are trying to answer in the next 10 slides
  - What happens to a conserved variable **over time** in a continuum?
- Answering this question allows us to study the time evolution of continua
- Mass, momentum, energy are examples of conserved variables
- Entropy in irreversible processes is an example of a variable that is not conserved

Conservation Principles

- Imagine a **control volume** ($\boldsymbol{V}$) surrounded by the **control surfaces** ($\boldsymbol{S}$)
- Denote $c = c(\boldsymbol{x}, t)$ as the concentration of a conserved scalar variable C per **unit mass** of the control volume:

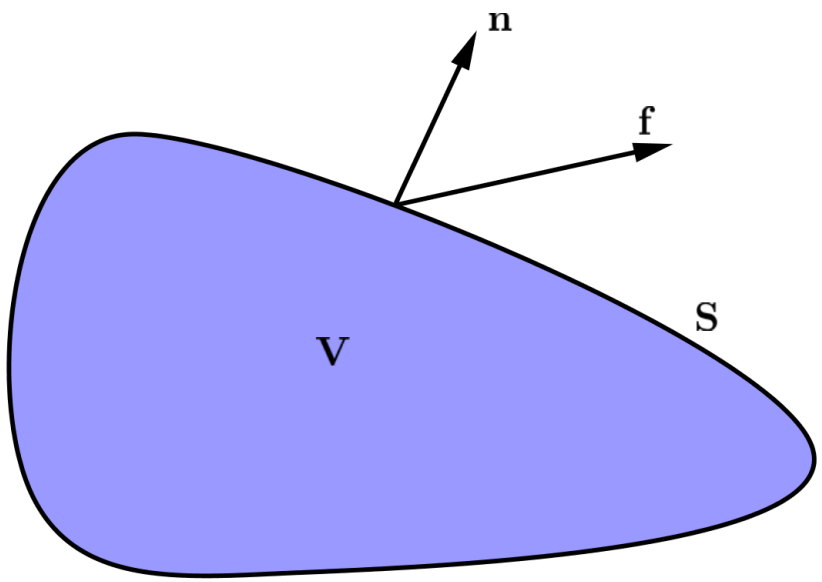
$$C = \int_V c(\boldsymbol{x}, t) dm = \int_V c(\boldsymbol{x}, t)\rho(\boldsymbol{x}, t) d\boldsymbol{x} = \int_V \phi(\boldsymbol{x}, t) d\boldsymbol{x}$$

- Define $\phi(\boldsymbol{x}, t) = \rho(\boldsymbol{x}, t)c(\boldsymbol{x}, t)$ to be the **volume** concentration of C, i.e. C per **unit volume**



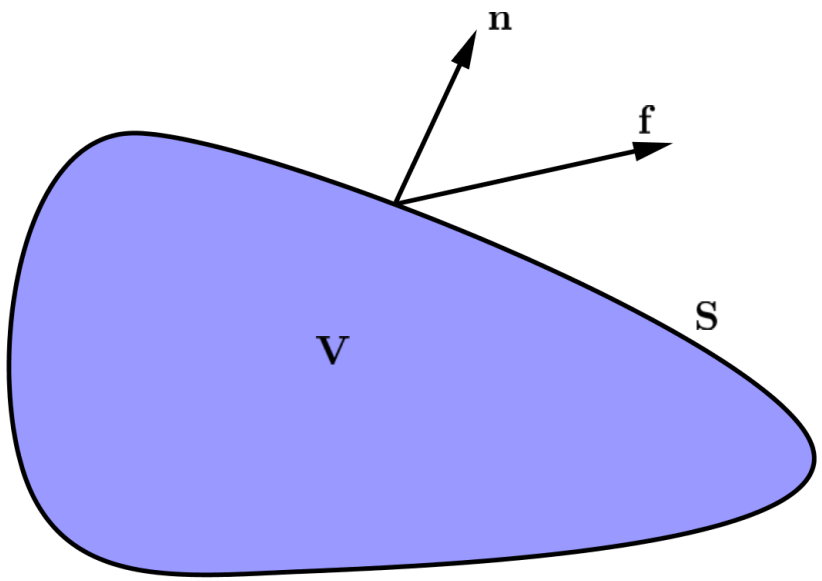
Conservation Principles

- Imagine a **control volume** ($\boldsymbol{V}$) surrounded by the **control surfaces** ($\boldsymbol{S}$)
- $\phi(\boldsymbol{x}, t) = \rho(\boldsymbol{x}, t)c(\boldsymbol{x}, t)$ is the **volume** concentration of C, i.e. C per **unit volume**
- Define **flux** of a variable to be the amount of the variable **per unit area and unit time** that enters or leaves $\boldsymbol{V}$ through $\boldsymbol{S}$
- The flux of a scalar variable is a vector, e.g., $\boldsymbol{f}$ in the figure



Conservation Principles

- Define **flux** of a variable to be the amount of the variable **per unit area and unit time** that enters or leaves $\boldsymbol{V}$ through $\boldsymbol{S}$
- The flux of a scalar variable is a vector, e.g., $\boldsymbol{f}$ in the figure
- The rate at which some amount of C enters or leaves the domain with flux $\boldsymbol{f} = \boldsymbol{f}(\boldsymbol{x}, t)$ is given by $\boldsymbol{f} \cdot \boldsymbol{n} ds$ for an infinitesimally small patch ds with a given normal $\boldsymbol{n}$

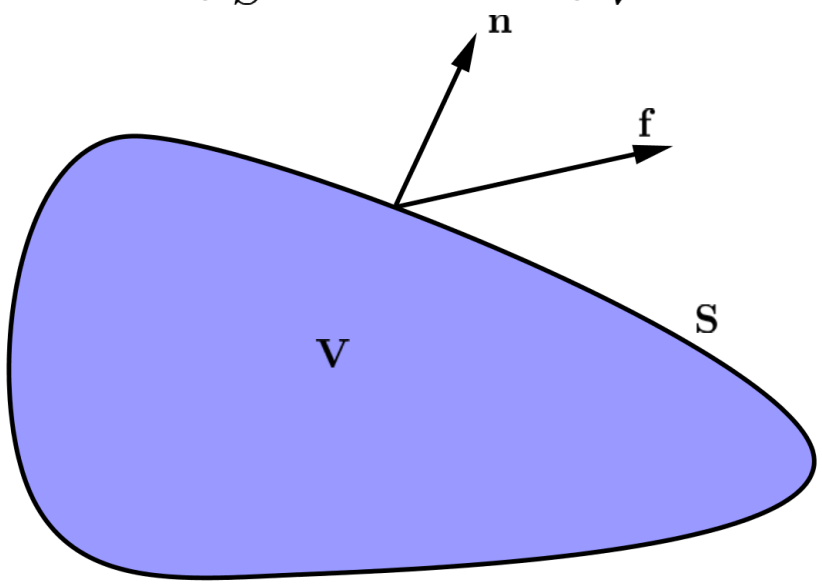


Conservation Principles - Integral form

- Conservation principle (the integral form):

the rate of change of a variable in the control volume V	$=$	the rate at which the variable enters or leaves the domain through the control surface S	$+$	the rate of generation or dissipation of the variable inside the control volume V

$$\frac{\partial}{\partial t}\int_V \phi(\boldsymbol{x},t)d\boldsymbol{x} = -\int_S \boldsymbol{f}\cdot\boldsymbol{n}ds + \int_V s(\boldsymbol{x},t)d\boldsymbol{x}$$



Conservation Principles

- Use Divergence theorem to convert the second surface integral to a volume integral as:

$$\int_S \boldsymbol{f} \cdot \boldsymbol{n} ds = \int_V (\nabla \cdot \boldsymbol{f}) d\boldsymbol{x}$$

- Simplify the conservation law as:

$$\frac{\partial}{\partial t} \int_V \phi(\boldsymbol{x}, t) d\boldsymbol{x} = \int_V (\nabla \cdot \boldsymbol{f}) d\boldsymbol{x} + \int_V s(\boldsymbol{x}, t) d\boldsymbol{x}$$

$$\int_V \left[\frac{\partial \phi(\boldsymbol{x}, t)}{\partial t} + \nabla \cdot \boldsymbol{f} - s(\boldsymbol{x}, t) \right] d\boldsymbol{x} = 0$$

- To obtain:

$$\frac{\partial \phi(\boldsymbol{x}, t)}{\partial t} + \nabla \cdot \boldsymbol{f}(\boldsymbol{x}, t) - s(\boldsymbol{x}, t) = 0$$

Conservation Principles - Differential Form

- This is the differential form of the conservation law:

$$\underbrace{\frac{\partial \phi(\boldsymbol{x},t)}{\partial t}}_{\text{rate of change of } \phi} = \underbrace{-\nabla \cdot \boldsymbol{f}(\boldsymbol{x},t)}_{\text{sink or source of the flux of } \phi} + \underbrace{s(\boldsymbol{x},t)}_{\text{rate of generation or dissipation of } \phi}$$

- This should make perfect sense. If the concentration ϕ is changing over time, it is either :
  - because of a sink or source in flux of ϕ, $\nabla \cdot \boldsymbol{f}(\boldsymbol{x},t)$ (remember the nature of divergence operator)
  - or, concentration is being generated or dissipated through a source of ϕ, i.e. $s(\boldsymbol{x},t)$
- The negative sign in $-\nabla \cdot \boldsymbol{f}(\boldsymbol{x},t)$ is due to the direction of the positive flux
  - For flux source $\nabla \cdot \boldsymbol{f}(\boldsymbol{x},t) > 0$, some amount of ϕ must leave the domain. Hence, it should decrease ϕ over time

Conservation Principles - Flux

- Think of the flux of a scalar field as a vector field
- Remember the divergence operator. A source/sink in this flux field requires some change in the underlying amount of ϕ in the domain
- The important question: How is the flux of ϕ, i.e., $f(\boldsymbol{x}, t)$ related to ϕ itself?
- It usually depends on the nature of ϕ, but generally there are two types of fluxes
  1. Diffusive flux: Transfer of ϕ driven by the difference in concentration of ϕ
  2. Advective flux: Transfer of ϕ because of an external velocity field
  3. Stress flux: You will see more about this later in the conservation of momentum where $\phi = \rho \boldsymbol{u}$

Conservation Principles – Diffusive Flux

- Diffusion: transfer of ϕ due to the difference in concentration of ϕ
- ϕ is transferred from the high concentration region to the low concentration region
- Remember the gradient operator. The gradient of ϕ , i.e., $\nabla\phi$, is a good measure for "difference in concentration"
- $\boldsymbol{f} = \boldsymbol{f}(\boldsymbol{x}, t) = \boldsymbol{D}\,\nabla\phi$ is the diffusive flux **vector**
- $\boldsymbol{D}$ is the diffusion coefficient; the larger the $\boldsymbol{D}$, the easier for ϕ to flow
- In general, $\boldsymbol{D}$ it is a second-order tensor, but it comes down to a scalar if there is no directional preference in diffusion

Conservation Principles – Advective Flux

- Advection: Transfer of ϕ via the flow velocity
- $\boldsymbol{f} = \boldsymbol{f}(\boldsymbol{x}, t) = \mathbf{u}\phi$ is the advective flux **vector**
- Note that the flux of a scalar is again a vector
- The larger the velocity, the larger the advective flux

Conservation Principles

- Revisit the conservation law of a conserved value ϕ

$$\underbrace{\frac{\partial \phi(\boldsymbol{x},t)}{\partial t}}_{\text{rate of change of } \phi} = \underbrace{-\nabla \cdot \boldsymbol{f}(\boldsymbol{x},t)}_{\text{sink or source of the flux of } \phi} + \underbrace{s(\boldsymbol{x},t)}_{\text{rate of generation or dissipation of } \phi}$$

- This equation gives us the time evolution of **any** conserved variable in **any** continuum
- The total flux vector is the combination of the advective and diffusive fluxes

$$\boldsymbol{f}(\boldsymbol{x},t) = \boldsymbol{u}(\boldsymbol{x},t)\phi(\boldsymbol{x},t) - \boldsymbol{D}(\boldsymbol{x},t)\nabla\phi(\boldsymbol{x},t)$$
$$\boldsymbol{f} = \boldsymbol{u}\phi - \boldsymbol{D}\nabla\phi$$

Conservation Principles

- Simplify the conservation law for the given flux vector:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\boldsymbol{u}\phi - \boldsymbol{D}\nabla\phi) - s = 0$$

- Assume $\nabla \cdot \boldsymbol{u} = 0$ (we shall say what this is later) and apply the following identity

$$\nabla \cdot (\boldsymbol{u}\phi) = \boldsymbol{u} \cdot \nabla\phi + \phi\nabla \cdot \boldsymbol{u}$$

- To get the Eulerian form of the conservation law:

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla\phi - \nabla \cdot (\boldsymbol{D}\nabla\phi) - s = 0$$

- Combine the first two terms via the material derivative formula to obtain the Lagrangian counterpart as:

$$\frac{D\phi}{Dt} - \nabla \cdot (\boldsymbol{D}\nabla\phi) - s = 0$$

Advection-Diffusion-Reaction Equation

- Consider the Eulerian form of the Advection-Diffusion-Reaction (ADR) equation

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi - \nabla \cdot (\boldsymbol{D} \nabla \phi) - s = 0$$

- Many engineering problems have this equation at their heart

- That is why we chose to study space-discretization methods, in the context of ADR equation

- Making further assumptions about this equation leads to important classes of PDEs:
  - **Elliptic** problems: Steady-state processes
  - **Parabolic** problems: diffusion-reaction dominant processes
  - **Hyperbolic** problems: advective dominant processes

Parabolic Equations

- Consider the Eulerian form of the conservation law

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi - \nabla \cdot (\boldsymbol{D} \nabla \phi) - s = 0$$

- **Parabolic** equations: If the advection term is negligible

$$\frac{\partial \phi}{\partial t} - \nabla \cdot (\boldsymbol{D} \nabla \phi) - s = 0$$

- Unsteady diffusion-driven processes such as mass diffusion or unsteady heat conduction
- In mass diffusion equation $\phi = c_i$ is the concentration of the i species
- In head conduction equation $\phi = T$ is the temperature.

Elliptic Equations

- Consider the Eulerian form of the conservation law

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi - \nabla \cdot (\boldsymbol{D} \nabla \phi) - s = 0$$

- **Elliptic** equations: Steady-state processes, where $\frac{\partial \boldsymbol{\phi}}{\partial t} = 0$, i.e.

$$\boldsymbol{u} \cdot \nabla \phi - \nabla \cdot (\boldsymbol{D} \nabla \phi) - s = 0$$

- If advection term is also negligible, and the diffusion tensor is $\boldsymbol{D} = d\boldsymbol{I}$ the Poisson equation is obtained

$$-d\nabla^2 \phi = -s$$

- The Poisson equation comes into play when solving the pressure equation in fluid flow problems

Hyperbolic Equations

- Consider the Eulerian form of the conservation law

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi - \nabla \cdot (\boldsymbol{D} \nabla \phi) - s = 0$$

- **Hyperbolic** equations: If the advection term is negligible

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi = s$$

- The mass conservation is an example where $\phi = \rho$ and $s = 0$, i.e., there is no mass sink or source in the field
- Another prominent example is the level-set equation which is obtained if ϕ is the level-set function

Weighted Residual Method (WRM)

- WRM helps in understanding the philosophy of many discretization methods
- For the sake of argument, consider the linear advection-diffusion equation described before in a domain $\boldsymbol{V}$

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\boldsymbol{u}\phi - \boldsymbol{D}\nabla\phi) - s = 0$$

- We are looking for finding $\phi = \phi(\boldsymbol{x}, t)$
- Assume that at some portion of the domain's boundary (Dirichlet portion, $S_D \subset S$) the value of the ϕ is known. i.e.,

$$\phi(\boldsymbol{x}, t) = \phi_D(\boldsymbol{x}),\ \forall \boldsymbol{x} \in S_D, \text{ and } \forall t \in (0, T)$$

- Assume that at some portion of the domain's boundary (Neumann portion, $S_N = S \setminus S_D$) the value of flux of ϕ is known. i.e.,

$$\boldsymbol{f}(\boldsymbol{x}, t) \cdot \boldsymbol{n} = g(\boldsymbol{x}),\ \forall \boldsymbol{x} \in S_N, \text{ and } \forall t \in (0, T)$$

Weighted Residual Method (WRM)

- Consider the residual $R(\bar{\phi})$ that results from an **approximate** solution $\bar{\phi}$

$$\mathcal{R}(\bar{\phi}) = \frac{\partial \bar{\phi}}{\partial t} + \nabla \cdot (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) - s$$

- $R(\bar{\phi})$ is a measure for solution accuracy, for the exact solution $R(\bar{\phi}) = 0$

- Imagine that we want to satisfy $R(\bar{\phi})$ in an integral sense, with more weight at some points in the region compared to others. Choose a function w (test function) from the space of admissible functions W. The condition for w is that it should vanish on the Dirichlet portion of the boundary, i.e., $w = 0$ for $x \in S_D$

$$\int_V w\mathcal{R}(\bar{\phi})d\boldsymbol{x} = 0, \qquad \forall w \in \mathcal{W}$$

$$\int_V w\left(\frac{\partial \bar{\phi}}{\partial t} + \nabla \cdot (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) - s\right)d\boldsymbol{x} = 0, \qquad \forall w \in \mathcal{W}$$

Weighted Residual Method (WRM)

- Applying the integration by parts on the flux term:

$$\int_V w\nabla\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})d\boldsymbol{x}=\int_V\nabla\cdot\big(w(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})\big)d\boldsymbol{x}-\int_V\nabla w\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})d\boldsymbol{x}$$

$$\int_V w\nabla\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})=\int_S w(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})\cdot\boldsymbol{n}\,ds-\int_V\nabla w\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})d\boldsymbol{x}$$

$$\int_V w\nabla\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})=\int_{S_N} w(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})\cdot\boldsymbol{n}\,ds-\int_V\nabla w\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})d\boldsymbol{x}$$

- And substitute back in the residual equation

$$\int_V(w\frac{\partial\bar{\phi}}{\partial t}-\nabla w\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})-ws)d\boldsymbol{x}+\int_{S_N} w(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})\cdot\boldsymbol{n}\,ds=0,\qquad\forall w\in\mathcal{W}$$

Weighted Residual Method (WRM)

- Apply the given information of the Neumann boundary conditions

$$\boldsymbol{f}(\boldsymbol{x},t)\cdot\boldsymbol{n}=(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})\cdot\boldsymbol{n}=g(\boldsymbol{x}),\ \forall\boldsymbol{x}\in S_N,\ \text{and}\ \forall t\in(0,T)$$

- This results in the weak/integral form of the conservation principle

$$\int_V(w\frac{\partial\bar{\phi}}{\partial t}-\nabla w\cdot(\boldsymbol{u}\bar{\phi}-\boldsymbol{D}\nabla\bar{\phi})-ws)d\boldsymbol{x}+\int_{S_N}w\ g\ ds=0,\qquad\forall w\in\mathcal{W}$$

Weighted Residual Method (WRM)

- Different choices of w results in the following famous techniques
- $w = 1$ (subdomain method) leads to the integral form or **Finite Volume** method

$$\int_V \frac{\partial \bar{\phi}}{\partial t} d\boldsymbol{x} + \int_{S_N} (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n}\, ds = \int_V s d\boldsymbol{x}$$

- $w = \delta(\boldsymbol{x} - \boldsymbol{x}_0)$ (collocation method) leads to the strong form used in **Finite Difference** method

$$\frac{\partial \bar{\phi}}{\partial t} + \nabla \cdot (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) = s \qquad \text{at } \boldsymbol{x} = \boldsymbol{x}_0$$

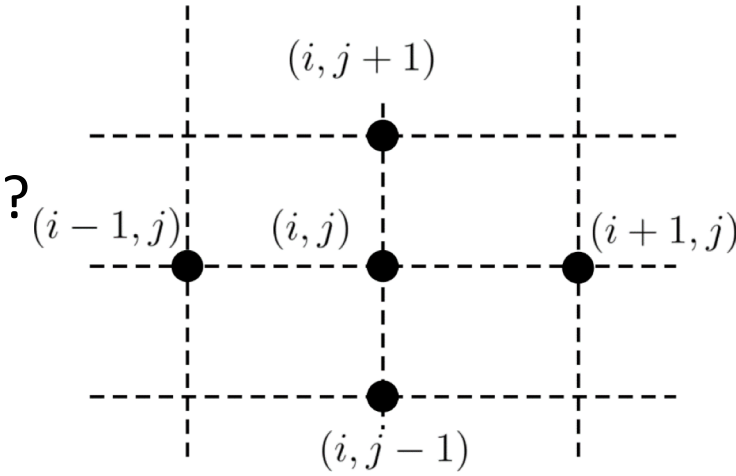
- $w = \psi_i$ in a **Finite Element** framework leads to the Galerkin method (more details will come up soon)

Finite Difference Method

- Most likely the oldest discretization technique for solving PDEs
- Remember, $w = \delta(\boldsymbol{x} - \boldsymbol{x}_0)$ leads to the discretization of the original PDE at collocation points

$$\frac{\partial \bar{\phi}}{\partial t} + \nabla \cdot (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) = s \qquad \text{at } \boldsymbol{x} = \boldsymbol{x}_0$$

- If the geometry of the domain is simple enough so you can have a uniform grid, then Finite difference becomes very attractive
- In 2D you solve the PDE on **grid points** (i, j)
- The DOFs are at defined on **vertices,** i.e. $\phi_{i,j}\,(t) = \phi(\boldsymbol{x}_{i,j}, t)$.
- The question is, how can you approximate the derivative terms?



Finite Difference Method

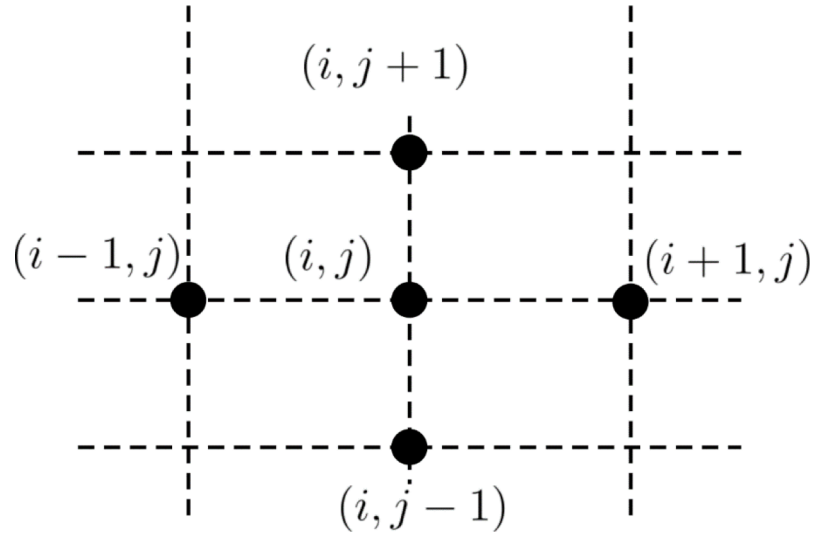
- The DOFs are at defined on **vertices,** i.e. $\phi_{i,j}\,(t) = \phi(\boldsymbol{x}_{i,j}, t)$
- Using Taylor expansion it can be shown that:

$$\frac{\partial \phi}{\partial x}|_{(i,j)} = \frac{\phi_{(i+1,j)}(t) - \phi_{(i-1,j)}(t)}{2\Delta x}$$

$$\frac{\partial \phi}{\partial y}|_{(i,j)} = \frac{\phi_{(i,j+1)}(t) - \phi_{(i,j-1)}(t)}{2\Delta y}$$

$$\frac{\partial^2 \phi}{\partial x^2}|_{(i,j)} = \frac{\phi_{(i+1,j)}(t) - 2\phi_{(i,j)}(t) + \phi_{(i-1,j)}(t)}{\Delta x^2}$$

$$\frac{\partial^2 \phi}{\partial y^2}|_{(i,j)} = \frac{\phi_{(i,j+1)}(t) - 2\phi_{(i,j)}(t) + \phi_{(i,j-1)}(t)}{\Delta y^2}$$



- Then you have a way to use the information of neighboring grid points to approximate spatial derivatives via second-order accurate schemes

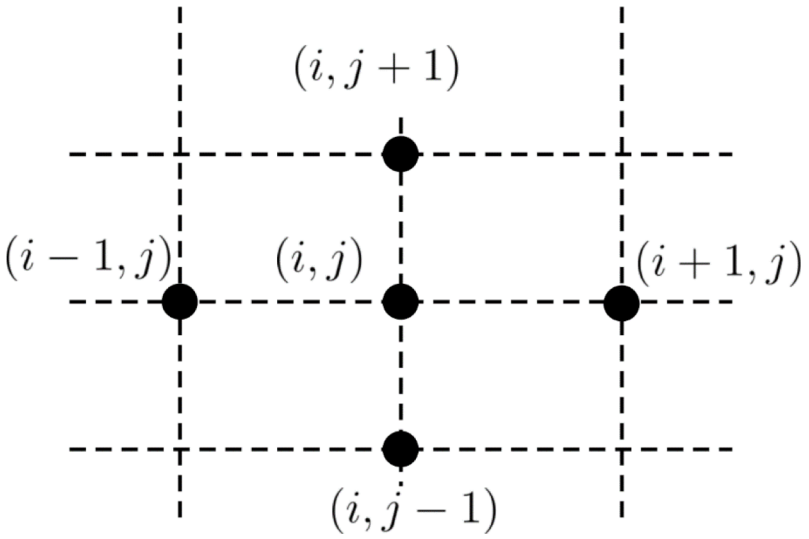
Finite Difference Method

- The DOFs are at defined on **vertices,** i.e. $\phi_{i,j}\,(t) = \phi(\boldsymbol{x}_{i,j}, t)$
- Similarly, it can be shown that the first-order approximation of the time-derivative is

$$\frac{\partial \phi_{(i,j)}(t)}{\partial t} = \frac{\phi_{(i,j)}(t^n) - \phi_{(i,j)}(t^{n-1})}{\Delta t}$$

- While the second-order approximation requires you to keep track of DOFs for two previous time-steps

$$\frac{\partial \phi_{(i,j)}(t)}{\partial t} = \frac{3\phi_{(i,j)}(t^n) - 4\phi_{(i,j)}(t^{n-1}) + \phi_{(i,j)}(t^{n-2})}{2\Delta t}$$



Finite Difference Method

- The DOFs are at defined on **vertices,** i.e. $\phi_{i,j}\,(t) = \phi(\boldsymbol{x}_{i,j}, t)$
- The conservation law with constant and diagonal diffusion $\boldsymbol{D}$, and $\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2}$

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla\phi - D\nabla^2\phi - s = 0$$

- The FD discretization of this equation is

$$\frac{\phi_{(i,j)}(t^n) - \phi_{(i,j)}(t^{n-1})}{\Delta t} + u_x \frac{\phi_{(i+1,j)}(t) - \phi_{(i-1,j)}(t)}{2\Delta x} + u_y \frac{\phi_{(i,j+1)}(t) - \phi_{(i,j-1)}(t)}{2\Delta y}$$
$$- D\left(\frac{\phi_{(i+1,j)}(t) - 2\phi_{(i,j)}(t) + \phi_{(i-1,j)}(t)}{\Delta x^2} + \frac{\phi_{(i,j+1)}(t) - 2\phi_{(i,j)}(t) + \phi_{(i,j-1)}(t)}{\Delta y^2}\right)$$
$$- s_{(i,j)} = 0$$

Finite Difference Method – Boundary Conditions

- For grid points on the boundary of the domain the algebraic equation is obtained from the boundary conditions rather than the PDE

- **Dirichlet Boundary:** $\phi_{(i,j)}(t) = \phi_D(\boldsymbol{x}_{i,j})$ is known at some boundary nodes
  - This equation itself is the algebraic equation for a Dirichlet node (i, j)

- **Neumann Boundary:** $(\nabla\phi) \cdot \boldsymbol{n}$ is known at some boundary nodes. If the point in on the **left** edge or the **right** edge of the domain, respectively the following second-order schemes hold:

$$(\nabla\phi) \cdot \boldsymbol{n} = h = \frac{3\phi_{(i,j)}(t) - 4\phi_{(i+1,j)}(t) + \phi_{(i+2,j)}(t)}{2\Delta x},$$

$$(\nabla\phi) \cdot \boldsymbol{n} = h = \frac{3\phi_{(i,j)}(t) - 4\phi_{(i-1,j)}(t) + \phi_{(i-2,j)}(t)}{2\Delta x}$$

Finite Difference Method – Final Thoughts

- The FD procedure results in an algebraic equation for each internal grid point
- Boundary conditions give algebraic equations for boundary points
- Assuming n grid points in the domain, we end up with n equations and n unknowns

- For the scalar conservation equation, we end up with a system of equations; sometimes, it is easy to form the system explicitly as $Ax = b$ and solve for x. Otherwise, matrix-free methods should be used

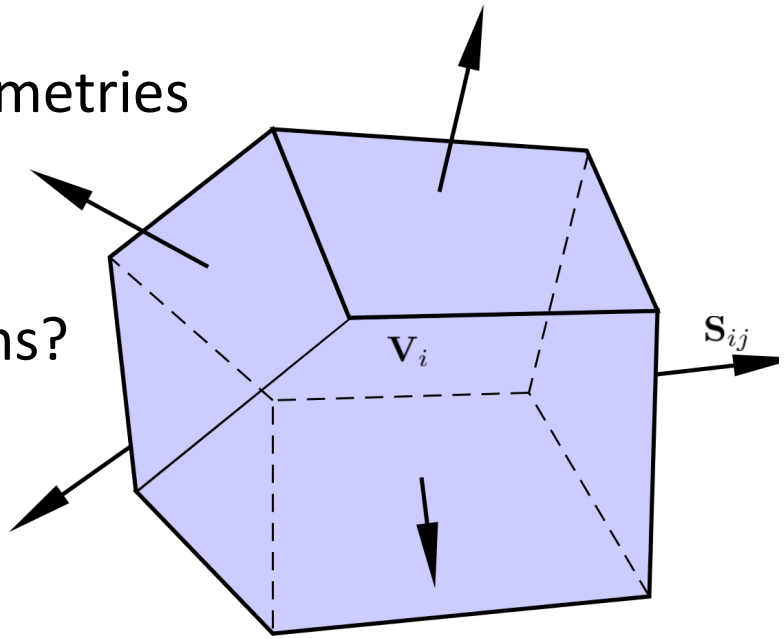
- Problem 1 walks you through the steps for solving this equation in 1D via FD method

Finite Volume Method

- Most widely exploited discretization technique in CFD
- Remember that the choice of $w = 1$ (subdomain method) in the WRM leads to the integral form or **Finite Volume** method

$$\int_V \frac{\partial \bar{\phi}}{\partial t} d\boldsymbol{x} + \int_{S_N} (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n}\, ds = \int_V s d\boldsymbol{x}$$

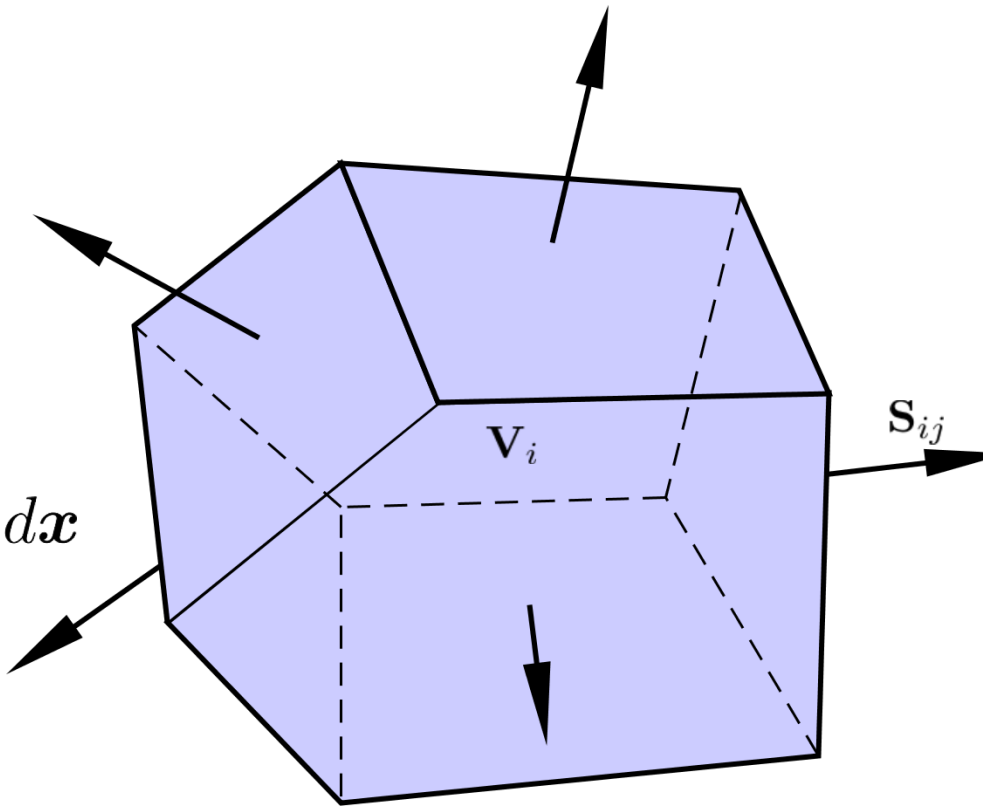
- This is very attractive because it works for complicated geometries
- The DOFs are (often) defined at **cell centers**
- It naturally allows for unstructured grids
- The question is, how can you approximate the integral terms?



Finite Volume Method

- Write the conservation law for cell i in the flux form

$$\frac{\partial}{\partial t}\int_{V_i}\phi(\boldsymbol{x},t)d\boldsymbol{x} = -\int_{S_i}\boldsymbol{f}\cdot\boldsymbol{n}ds + \int_{V_i}s(\boldsymbol{x},t)d\boldsymbol{x}$$



- The cell V_i bounded by the control surface S_i. The control surface S_i consists of flat surfaces such as $\mathbf{S}_{ij}$ with known surface area S_{ij} and surface normal vector $\boldsymbol{n}_{ij}$ such that $\mathbf{S}_{ij} = S_{ij}\boldsymbol{n}_{ij}$

Finite Volume Method

- The conservation law for cell i

$$\frac{\partial}{\partial t}\int_{V_i}\phi(\boldsymbol{x},t)d\boldsymbol{x} = -\int_{S_i}\boldsymbol{f}\cdot\boldsymbol{n}ds + \int_{V_i}s(\boldsymbol{x},t)d\boldsymbol{x}$$

- The DOFs are (often) defined at **cell centers**, e.g., $\phi_i(t)$ is the average value of ϕ inside the cell i. i.e.

$$\phi_i(t) = \frac{1}{|V_i|}\int_{V_i}\phi(\boldsymbol{x},t)dV$$

- Similarly, the source term in the cell can be approximated as the average value of the cell, i.e.

$$s_i(t) = \frac{1}{|V_i|}\int_{V_i}s(\boldsymbol{x},t)dV$$

- If computational cells are polyhedral, i.e., the control surface consists of flat surfaces:

$$\int_{S_i}\boldsymbol{f}\cdot\boldsymbol{n}ds = \sum_{S_{ij}}\boldsymbol{f}\cdot\boldsymbol{S_{ij}}$$

Finite Volume Method

- The discrete form of the conservation law for cell i becomes

$$\frac{\partial}{\partial t}\int_{V_i}\phi(\boldsymbol{x},t)d\boldsymbol{x} = -\int_{S_i}\boldsymbol{f}\cdot\boldsymbol{n}ds + \int_{V_i}s(\boldsymbol{x},t)d\boldsymbol{x}$$

$$|V_i|\frac{\partial\phi_i(t)}{\partial t} = -\sum_{S_{ij}}\boldsymbol{f}\cdot\boldsymbol{S_{ij}} + |V_i|s_i(t)$$

$$|V_i|\frac{\partial\phi_i(t)}{\partial t} = -\sum_{S_{ij}}(\boldsymbol{u}\phi - \boldsymbol{D}\nabla\phi)\cdot\boldsymbol{S_{ij}} + |V_i|s_i(t)$$

$$|V_i|\frac{\partial\phi_i(t)}{\partial t} = -\underbrace{\sum_{S_{ij}}(\boldsymbol{u}\phi)\cdot\boldsymbol{S_{ij}}}_{\text{advctive flux}} + \underbrace{\sum_{S_{ij}}(\boldsymbol{D}\nabla\phi)\cdot\boldsymbol{S_{ij}}}_{\text{diffusive flux}} + |V_i|s_i(t)$$

- We just need to find out how to evaluate the advective and diffusive flux at S_{ij}

Finite Volume Method- Advective Flux

- The question is how to evaluate the $\boldsymbol{u\phi \cdot S_{ij}}$term in the following?

$$|V_i|\frac{\partial \phi_i(t)}{\partial t} = -\underbrace{\sum_{S_{ij}}(\boldsymbol{u}\phi)\cdot \boldsymbol{S_{ij}}}_{\text{advctive flux}} + \underbrace{\sum_{S_{ij}}(\boldsymbol{D}\nabla\phi)\cdot \boldsymbol{S_{ij}}}_{\text{diffusive flux}} + |V_i|s_i(t)$$

- This is similar as asking what is ϕ_{ij} in the following equation at the cell faces S_{ij}

$$\sum_{S_{ij}}(\boldsymbol{u}\phi)\cdot \boldsymbol{S_{ij}} = \sum_{S_{ij}}(\boldsymbol{u}\cdot \boldsymbol{S_{ij}})\phi_{ij} = \sum_{S_{ij}} F_{ij}\ \phi_{ij},$$

Central Difference: a second-order accurate scheme that linearly interpolates between ϕ_i and ϕ_j, i.e.

$$\phi_{ij} = f_x\phi_i + (1-f_x)\phi_j$$

  - $f_x = d_i/d_{ij}$, and d_{ij} is the distance between the center of cell i and the center of cell j, and d_i is the distance between the center of cell i and the face between cell i and cell j.

Finite Volume Method- Advective Flux

2. Upwind Difference: a first-order accurate scheme that adds numerical dissipation and increases stability by assuming that ϕ_{ij} is either ϕ_i or ϕ_j, depending on the velocity direction at the face ij

$$\phi_{ij} = \begin{cases} \phi_i & \text{if } F_{ij} = \boldsymbol{u} \cdot \boldsymbol{S_{ij}} \geq 0 \\ \phi_j & \text{if } F_{ij} = \boldsymbol{u} \cdot \boldsymbol{S_{ij}} < 0 \end{cases}$$

3. Blended Difference: a combination of the central differencing (CD) and upwind differencing (UD) schemes. It can be as dissipative (stable) as UD or as accurate as CD depending on γ

$$\phi_{ij} = \gamma \phi_{UD} + (1 - \gamma)\phi_{CD}$$

Finite Volume Method- Diffusive Flux

- The question is how to evaluate the diffusive flux $(\boldsymbol{D}\nabla\boldsymbol{\phi})_{ij}$ term at the cell faces S_{ij}?

$$|V_i|\frac{\partial \phi_i(t)}{\partial t} = -\underbrace{\sum_{S_{ij}}(\boldsymbol{u}\phi)\cdot\boldsymbol{S_{ij}}}_{\text{advctive flux}} + \underbrace{\sum_{S_{ij}}(\boldsymbol{D}\nabla\phi)\cdot\boldsymbol{S_{ij}}}_{\text{diffusive flux}} + |V_i|s_i(t)$$

- If the face normal is aligned with the line connecting cell i and cell j, which is the case in orthogonal meshes the following is an accurate approximation

$$\sum_{S_{ij}}(\boldsymbol{D}\nabla\phi)_{ij}\cdot\boldsymbol{S_{ij}} = \sum_{S_{ij}}(\boldsymbol{D}_{ij}\nabla\phi_{ij}\cdot\boldsymbol{S_{ij}}) = \sum_{S_{ij}} D_{ij}S_{ij}\frac{\phi_i-\phi_j}{d_{ij}}$$

Finite Volume Method – Boundary Conditions

- **Dirichlet boundary conditions ($\phi|_{S_D} = \phi_D$)**
  - This is similar as a given advective flux $\boldsymbol{u\phi \cdot S}\,|_{S_D} = \boldsymbol{f}_{adv}$
  - ϕ_{ij} in the advective flux term is substituted with ϕ_D for any face on the Dirichlet part of the boundary

$$\sum_{S_{ij}} (\boldsymbol{u}\phi) \cdot \boldsymbol{S_{ij}} = \sum_{S_{ij}} (\boldsymbol{u} \cdot \boldsymbol{S_{ij}}) \phi_{ij} = \sum_{S_{ij}} F_{ij}\ \phi_{ij}$$

  - The gradient term associated with the diffusive flux i.e. $S_{ij}\,\frac{(\phi_i - \phi_j)}{d_{ij}}$ in

$$\sum_{S_{ij}} (\boldsymbol{D}\nabla\phi)_{ij} \cdot \boldsymbol{S_{ij}} = \sum_{S_{ij}} (\boldsymbol{D}_{ij}\nabla\phi_{ij} \cdot \boldsymbol{S_{ij}}) = \sum_{S_{ij}} D_{ij} S_{ij} \frac{\phi_i - \phi_j}{d_{ij}}$$

  is substituted with $S_f\,\frac{(\phi_i - \phi_D)}{d}$, d is the distance between the boundary face and its cell center, and S_f is the magnitude area of the boundary face

Finite Volume Method – Boundary Conditions

- **Neumann boundary conditions ($(\nabla \cdot \phi) \cdot \boldsymbol{n} = g_{\phi}$)**
  - This is similar as a given diffusive flux $\boldsymbol{D}\nabla\phi \cdot \boldsymbol{S}\,|_{S_N} = \boldsymbol{f}_{diff}$
  - ϕ_{ij} in the advective flux term

$$\sum_{S_{ij}} (\boldsymbol{u}\phi) \cdot \boldsymbol{S_{ij}} = \sum_{S_{ij}} (\boldsymbol{u} \cdot \boldsymbol{S_{ij}})\phi_{ij} = \sum_{S_{ij}} F_{ij}\ \phi_{ij}$$

  is interpolated from the face gradient via $\phi_f = \phi_i + \boldsymbol{d} \cdot \nabla\phi_f = \phi_i + d\ g_{\phi}$ for any face on the Neumann part of the boundary. d is the distance between the boundary face and its cell center

  - The gradient term associated with the diffusive flux i.e., $\nabla\phi_f \cdot \boldsymbol{S}_f$ in

$$\sum_{S_{ij}} (\boldsymbol{D}\nabla\phi)_{ij} \cdot \boldsymbol{S_{ij}} = \sum_{S_{ij}} (\boldsymbol{D}_{ij}\nabla\phi_{ij} \cdot \boldsymbol{S_{ij}}) = \sum_{S_{ij}} D_{ij} S_{ij} \frac{\phi_i - \phi_j}{d_{ij}}$$

  is substituted with $g_{\phi}\ S_f$

Finite Volume Method – Final Thoughts

- The FV procedure results in an algebraic equation for each internal **cell**
- Boundary conditions are incorporated into the governing equations of the cell with a boundary face

- Assuming n cells in the domain, we end up with n equations and n unknowns

- For the scalar conservation equation, we end up with a system of equations; sometimes, it is easy to form the system explicitly as $Ax = b$ and solve for x. Otherwise, matrix-free methods should be used

- Problem 2 walks you through the steps for solving this equation in 1D via FV method

Smoothed Particle Hydrodynamics (SPH)

- Contrary to FD and Eulerian methods where degrees of freedom are fixed in SPH the degrees of freedom are moving with the flow velocity
- The idea is to define degrees of freedom at imaginary particles (or markers, to be precise)
- If the approximation of functions f at any point $\boldsymbol{r}_i$ in space is needed, we use the weighted average of f defined at degrees of freedom, i.e., $f(\boldsymbol{r}_j)$s for particle j

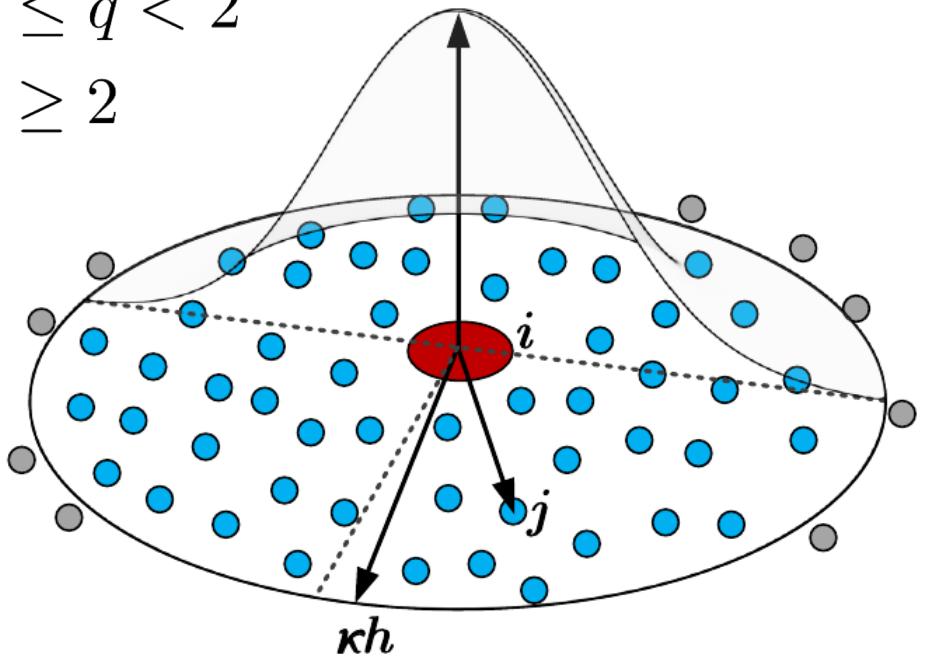
$$f(\mathbf{r}_i) \approx \langle f \rangle_i = \sum_{j \in \mathcal{S}(i)} \frac{m_j}{\rho_j} f(\mathbf{r}_j) W_{ij}$$

- m_j and ρ_j are the mass and density of the marker j, and $V_j = m_j/\rho_j$
- W_{ij} is the weight function. The closer $\boldsymbol{r}_j$to $\boldsymbol{r}_i$ the larger W_{ij}
- If marker j is too far away from $\boldsymbol{r}_i$then it should not contribute much, so W_{ij}should be very small. $S(i)$ is the set of particles that in fact have a significant effect in this summation

Smoothed Particle Hydrodynamics (SPH)

- $q = r/h$ and $r = |\boldsymbol{r}_i - \boldsymbol{r}_j|$.
- h is a characteristic length, and it usually is the initial distance of markers
- An example of the weight (kernel) function

$$W(|\mathbf{r}|, h) = \frac{5}{14\pi h^3} \times \begin{cases} (2-q)^3 - 4(1-q)^3, & 0 \le q < 1 \\ (2-q)^3, & 1 \le q < 2 \\ 0, & q \ge 2 \end{cases}$$



Smoothed Particle Hydrodynamics (SPH)

- Spatial derivatives can be calculated by taking the derivative of the weight function
- Amongst various schemes to approximate, these are examples for approximating gradient and Laplacian of a function

$$\nabla f(\mathbf{r}_i) = \langle \nabla f \rangle_i = \sum_{j \in \mathcal{S}(i)} V_j (f_j - f_i) \nabla_i W_{ij}$$

$$\nabla^2 f(\mathbf{r}_i) = \langle \nabla^2 f \rangle_i = 2 \sum_{j \in \mathcal{S}(i)} V_j (\mathbf{e}_{ij} \cdot \nabla_i W_{ij}) \frac{f_i - f_j}{|\mathbf{r}_{ij}|}$$

- You see that we require the derivative of the kernel function, if $\boldsymbol{e_{ij}} = \frac{\boldsymbol{r_{ij}}}{|\boldsymbol{r_{ij}}|}$

$$\nabla_i W_{ij} = \frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|} \frac{\partial W}{\partial q} \frac{\partial q}{\partial |\mathbf{r}_{ij}|}\bigg|_{i,j} = \frac{-15 \mathbf{r}_{ij}}{14 \pi h^5 q} \times \begin{cases} (2-q)^2 - 4(1-q)^2, & 0 \le q < 1 \\ (2-q)^2, & 1 \le q < 2 \\ 0, & q \ge 2 \end{cases}$$

Smoothed Particle Hydrodynamics– Final Thoughts

- The SPH procedure results in an algebraic equation for each internal **marker/particle**
- Implementing boundary conditions is still a matter of research

- Assuming n markers in the domain, we end up with n equations and n unknowns

- For the scalar conservation equation, we end up with a system of equations; sometimes it is easy to form the system explicitly as $Ax = b$ and solve for x. Otherwise, matrix-free methods should be used

- Problem 4 walks you through the steps for solving this equation in 1D via SPH method

Finite Element Method

- Start from the week form of the equations:

$$\int_V (w\frac{\partial \bar{\phi}}{\partial t} - \nabla w \cdot (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) - ws)d\boldsymbol{x} + \int_{S_N} w\ g\ ds = 0, \qquad \forall w \in \mathcal{W}$$

- Remember that this equation was derived after integration by part and applying the Neumann boundary condition
- The idea behind FEM is to :
  - choose a suitable weight function w,
  - form the above integral equation for each node i
  - Solve the system of equations to find nodal degrees of freedom $\phi_i(t) = \phi(\boldsymbol{x}_i, t)$
- Amongst different choices of weight function, we limit our discussion to the Galerkin method

Finite Element Method - Introduction

- Start from a simple idea, separation of space and time functionality in $\phi(\boldsymbol{x}, t)$ by :

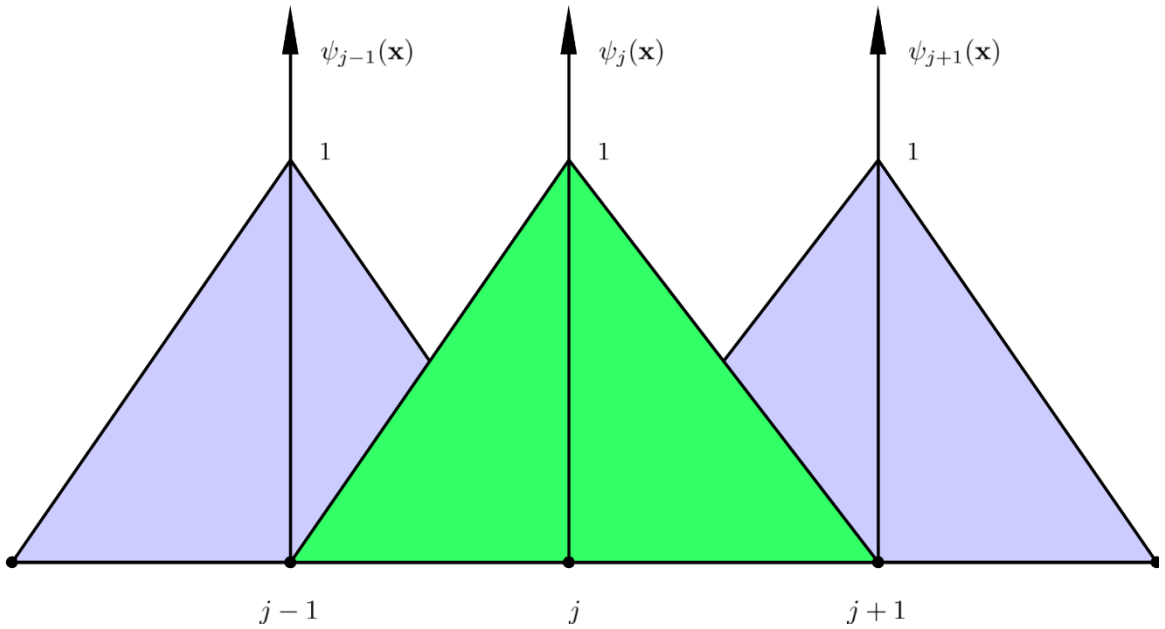
$$\bar{\phi}(\boldsymbol{x}, t) = \sum_j \phi_j(t)\psi_j(\boldsymbol{x})$$

  - $\bar{\phi}(\boldsymbol{x}, t)$ is the approximate solution to $\phi(\boldsymbol{x}, t)$
  - $\phi_j(t)$ are nodal values of ϕ at the nodes $j = 1,2, \dots, \mathrm{n}$ and are only a function of time
  - $\psi_j(\boldsymbol{x})$ are the nodal interpolation/shape/basis functions at the nodes $j = 1,2, \dots, \mathrm{n}$ are only a function of space

- The idea is to approximate the solution in space by interpolating the nodal values ϕ_j
- However, the nodal values $\phi_j(t)$ are unknown at this point

Finite Element Method – Basis functions

- Basis functions $\psi_j(\boldsymbol{x})$ in the finite element interpolation formula have certain properties
  - it is 1 if evaluated at the node j and is 0 if evaluated at any other node $i \neq j$, i.e. $\psi_j(\boldsymbol{x}_j) = 1$ and $\psi_j(\boldsymbol{x}_i) = 0$ if $i \neq j$
  - $\psi_j(\boldsymbol{x})$ are special functions in that they are non-zero only in the cells surrounding the node/vertex j; elsewhere they are 0
  - in the 1D case:



Finite Element Method –Derivatives

- The separation of time and space in $\phi(\boldsymbol{x}, t)$:

$$\bar{\phi}(\boldsymbol{x}, t) = \sum_j \phi_j(t)\psi_j(\boldsymbol{x})$$

- Allows to calculate the gradient and Laplacian of the ϕas follow:

$$\nabla\bar{\phi}(\boldsymbol{x}, t) = \sum_j \phi_j(t)\nabla\psi_j(\boldsymbol{x}),$$

$$\nabla^2\bar{\phi}(\boldsymbol{x}, t) = \sum_j \phi_j(t)\nabla^2\psi_j(\boldsymbol{x})$$

- Time-derivative can be calculated from

$$\frac{\partial\bar{\phi}(\boldsymbol{x}, t)}{\partial t} = \sum_j \frac{\partial\phi_j(t)}{\partial t}\psi_j(\boldsymbol{x})$$

Finite Element Method –Galerkin Method

- The finite element formulation can be applied to any PDE, but here we are interested in the weak/integral form of the conservation principle, i.e.

$$\int_V (w\frac{\partial \bar{\phi}}{\partial t} - \nabla w \cdot (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) - ws)d\boldsymbol{x} + \int_{S_N} w\ g\ ds = 0, \qquad \forall w \in \mathcal{W}$$

- This equation holds for any point in the domain
- We form one discrete equation for each node i in the mesh
- In the standard continuous-Galerkin method, the weight/test function for vertex i is the same as the interpolation function, i.e., $w = \psi_i$
- We need to incorporate the derivative formulations discussed before

Finite Element Method –Galerkin Method

- It follows that for any node i with $w = \psi_i$

$$\sum_j \int_V \left(\psi_i(\boldsymbol{x}) \frac{\partial \phi_j(t)}{\partial t} \psi_j(\boldsymbol{x}) - \nabla \psi_i(\boldsymbol{x}) \cdot (\boldsymbol{u}_j \phi_j(t) \psi_j(\boldsymbol{x})(\boldsymbol{x}) - \boldsymbol{D} \phi_j(t) \nabla \psi_j(\boldsymbol{x})) \right) d\boldsymbol{x}$$

$$- \int_V \psi_i(\boldsymbol{x}) s(\boldsymbol{x}) d\boldsymbol{x} + \int_{S_N} \psi_i(\boldsymbol{x}) (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n} \, ds = 0$$

- Since integration is over space, the time-functions is taken out of the integral

$$\sum_j \left(\int_V \psi_i \psi_j d\boldsymbol{x} \right) \frac{\partial \phi_j(t)}{\partial t} - \sum_j \left(\int_V \nabla \psi_i \cdot (\boldsymbol{u}_j \psi_j - \boldsymbol{D} \nabla \psi_j) d\boldsymbol{x} \right) \phi_j(t)$$

$$- \int_V \psi_i s d\boldsymbol{x} + \int_{S_N} \psi_i (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n} \, ds = 0$$

Finite Element Method –Galerkin Method

- Note that we assumed that on the Neumann part of the boundary the total flux is known, i.e., $\boldsymbol{f} = (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n}$
- See the details in the notes provided, if only the diffusive flux is known on the Neumann part of the boundary, i.e., $\boldsymbol{f} = (-\boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n}$
- Define the following terms:

$$m_{ij} = \int_V \psi_i \psi_j d\boldsymbol{x}$$

$$\boldsymbol{c}_{ij} = \int_V \psi_i \nabla \psi_j d\boldsymbol{x}$$

$$d_{ij} = \int_V \nabla\psi_i \cdot (\boldsymbol{D}\nabla\psi_j) d\boldsymbol{x}$$

$$r_i^{full} = \int_V \psi_i s d\boldsymbol{x} - \int_{S_N} \psi_i (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n}\, ds$$

Finite Element Method – Element Matrices

- You may think about individual m_{ij}, $\boldsymbol{c}_{ij}$, d_{ij} as respectively mass, convection, and diffusion between two nodes. They are large for adjacent nodes

- r_i^{full} is the load associated with a node i

- How does one calculate these integrals over the whole domain's volume? Fortunately, this is not necessary as the support of the shape functions is limited to **only adjacent elements** of each node, elsewhere ψ_i associated with node i is zero

Finite Element Method – Element Matrices

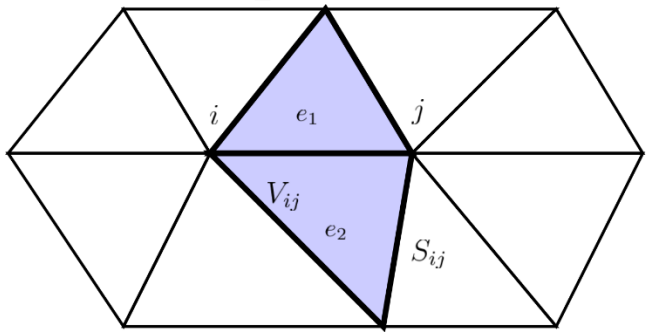
- For example, in 2D, calculation of m_{ij} comes down to calculating the integral over the **two elements** e_1 and e_2 . Over other elements either $\psi_i = 0$ or $\psi_j = 0$

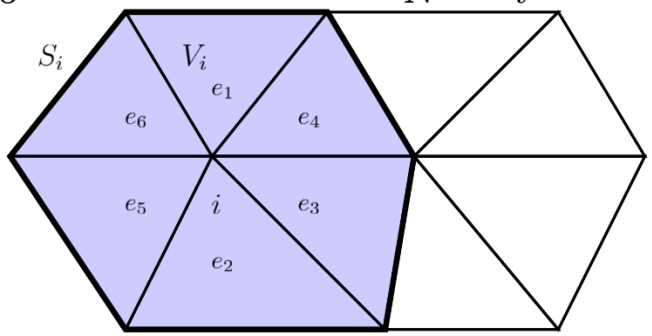
$$m_{ij} = \int_V \psi_i \psi_j d\boldsymbol{x} = \int_{e_1} \psi_i \psi_j d\boldsymbol{x} + \int_{e_2} \psi_i \psi_j d\boldsymbol{x}$$

- Calculating vector quantities such as load vector r_i^{full} comes down to integration over adjacent elements to node i and over surfaces shared with the exterior Neumann boundary

$$r_i = \int_V \psi_i s d\boldsymbol{x} - \int_{S_N} \psi_i (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n}\, ds$$

$$\int_{e_1} \psi_i s V + \int_{e_2} \psi_i s V + \cdots + \int_{e_6} \psi_i s V + - \int_{S_N \cap S_i} \psi_i (\boldsymbol{u}\bar{\phi} - \boldsymbol{D}\nabla\bar{\phi}) \cdot \boldsymbol{n}\, ds$$





Finite Element Method – Global Matrices

- Note that calculating m_{ij}, $\boldsymbol{c}_{ij}$, and d_{ij} boiled down to calculating these quantities over each element
- In practice, we calculate these quantities over each element and then assemble the global matrices
- For instance, we calculate the following mass matrix integral over element e_k. Indices 1,2,…,n are the local index of the nodes of the element k

$$\boldsymbol{M}_{e_k} = \int_{e_k} \begin{bmatrix} \psi_1\psi_1 & \psi_1\psi_2 & \cdots \\ \psi_2\psi_1 & \psi_1\psi_2 & \cdots \\ \psi_3\psi_1 & \psi_1\psi_2 & \cdots \\ \vdots & \vdots & \vdots \\ \psi_{e_n}\psi_1 & \psi_{e_n}\psi_2 & \cdots \end{bmatrix} d\boldsymbol{x}$$

- $\int_{e_k} \psi_i \, \psi_j$ part contribute to the $m_{g_i \, g_j}$ element in the global mass matrix. g_i and g_j are the global index of the local node i and j, respectively

Finite Element Method – Global Matrices

- Similarly, the following are the element's advection and diffusion matrices

$$\boldsymbol{C}_{e_k} = \int_{e_k} \begin{bmatrix} \psi_1 \nabla \psi_1 & \psi_1 \nabla \psi_2 & \cdots \\ \psi_2 \nabla \psi_1 & \psi_1 \nabla \psi_2 & \cdots \\ \psi_3 \nabla \psi_1 & \psi_1 \nabla \psi_2 & \cdots \\ \vdots & \vdots & \vdots \\ \psi_{e_n} \nabla \psi_1 & \psi_{e_n} \nabla \psi_2 & \cdots \end{bmatrix} d\boldsymbol{x}$$

$$\boldsymbol{D}_{e_k} = \int_{e_k} \begin{bmatrix} \nabla \psi_1 \cdot (\boldsymbol{D} \nabla \psi_1) & \nabla \psi_1 \cdot (\boldsymbol{D} \nabla \psi_2) & \cdots \\ \nabla \psi_2 \cdot (\boldsymbol{D} \nabla \psi_1) & \nabla \psi_1 \cdot (\boldsymbol{D} \nabla \psi_2) & \cdots \\ \nabla \psi_3 \cdot (\boldsymbol{D} \nabla \psi_1) & \nabla \psi_1 \cdot (\boldsymbol{D} \nabla \psi_2) & \cdots \\ \vdots & \vdots & \vdots \\ \nabla \psi_{e_n} \cdot (\boldsymbol{D} \nabla \psi_1) & \nabla \psi_{e_n} \cdot (\boldsymbol{D} \nabla \psi_2) & \cdots \end{bmatrix} d\boldsymbol{x}$$

Finite Element Method – Global Matrices

- Finally, the elements load vector is calculated via

$$\boldsymbol{r}_{e_k}^{full} = \int_{e_k} \begin{bmatrix} \psi_1 s \\ \psi_2 s \\ \psi_3 s \\ \vdots \\ \psi_{e_n} s \end{bmatrix} d\boldsymbol{x} - \int_{S_N \cap S_{e_k}} \begin{bmatrix} \psi_1 (\boldsymbol{u}_1 \phi_1 - \boldsymbol{D} \nabla \phi_1) \cdot \boldsymbol{n} \\ \psi_2 (\boldsymbol{u}_1 \phi_2 - \boldsymbol{D} \nabla \phi_2) \cdot \boldsymbol{n} \\ \psi_3 (\boldsymbol{u}_1 \phi_3 - \boldsymbol{D} \nabla \phi_3) \cdot \boldsymbol{n} \\ \vdots \\ \psi_{e_n} (\boldsymbol{u}_{e_n} \phi_{e_n} - \boldsymbol{D} \nabla \phi_{e_n}) \cdot \boldsymbol{n} \end{bmatrix} ds$$

- similar mapping from local indices to global indices are performed

Finite Element Method – System Matrices

- We can write the discrete equation for node i as follows:

$$\sum_j m_{ij} \frac{\partial \phi_j(t)}{\partial t} + \sum_j \Big(- \boldsymbol{u}_j \cdot \boldsymbol{c}_{ij} + d_{ij} \Big) \phi_j(t) - r_i^{full} = 0$$

- If we assemble the equations for all the nodes in the domain, the system level equation can be written as follow:

$$\boldsymbol{M} \frac{\partial \phi(t)}{\partial t} - \sum_j \boldsymbol{K} \phi(t) - \boldsymbol{r} = 0$$

- $\boldsymbol{M}$ is the system mass matrix assembled from elements' m_{ij}, $\boldsymbol{K}$ is the stiffness matrix assembled from the elements' stiffnesses $k_{ij} = \boldsymbol{u}_j \cdot \boldsymbol{c}_{ij} - d_{ij}$, and $\boldsymbol{r}$ is the load vector obtained from elements' loads r_i

Finite Element Method – Boundary Conditions

- Note that the Neumann boundary conditions is taken into the weak form already

- Dirichlet boundary condition can be imposed in two ways:
  - **Strong enforcement**: We replace the algebraic equation of node i with $\phi_i = \phi_D(\boldsymbol{x}_i)$ using the information of the Dirichlet boundary condition

  - **Weak enforcement**: Use the Neumann integral in the equations to enforce Dirichlet BC. Replace the term $(\boldsymbol{u}\bar{\phi} - D\ \nabla\bar{\phi}) \cdot \boldsymbol{n}$ in the surface integral part of the load vector with $\boldsymbol{u}\phi_D(\boldsymbol{x}_i)$. Note that via this method we satisfy the boundary conditions only in an integral (weak) sense

Finite Element Method – Numerical Integration

- We stated that elements matrices should be calculated before reconstructing the global mass, and stiffness matrices along with the load vector
- How to numerically calculate these integrals over an arbitrary-shaped element?
- In the following, you will see how this is done for iso-parametric elements
- In Iso-parametric elements the same shape functions are used to interpolate the coordinates of elements, remember $\phi(\boldsymbol{x}, t) = \sum_j \phi_j(t)\psi_j(\boldsymbol{x})$
- Here, we use the same functions in terms of the natural coordinates to express the coordinates of elements as : $\boldsymbol{x}(\boldsymbol{\zeta}, t) = \sum_j \boldsymbol{x}_j(t)\psi_j(\boldsymbol{\zeta})$

Finite Element Method – Iso-parametric elements

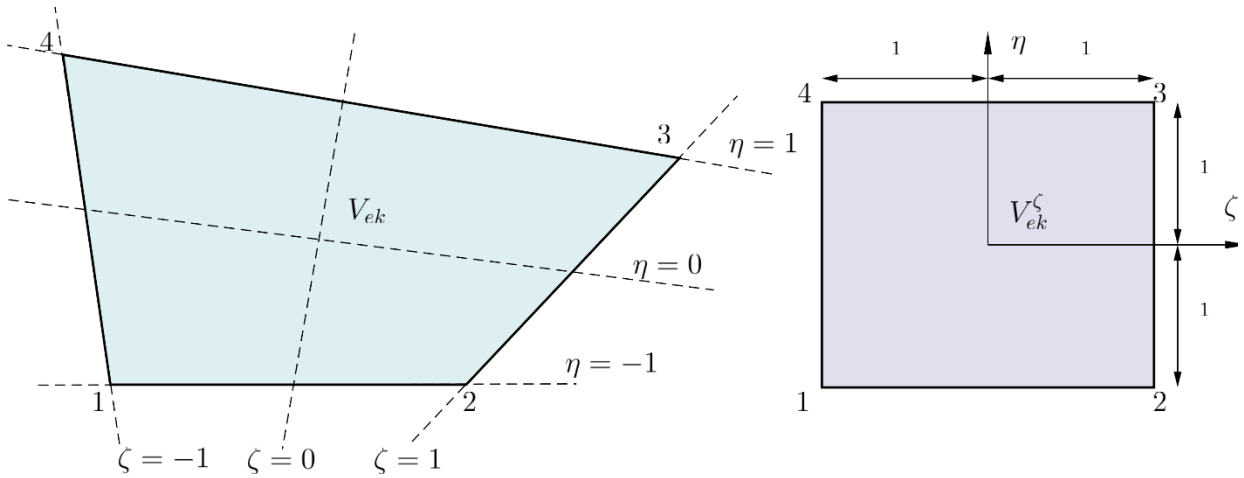
- A typical bi-linear four-node element for 2D problems:

$$\psi_1(\boldsymbol{\zeta}) = \psi_1(\zeta,\eta) = \frac{1}{4}(1-\zeta)(1-\eta)$$

$$\psi_2(\boldsymbol{\zeta}) = \psi_2(\zeta,\eta) = \frac{1}{4}(1+\zeta)(1-\eta)$$

$$\psi_3(\boldsymbol{\zeta}) = \psi_3(\zeta,\eta) = \frac{1}{4}(1+\zeta)(1+\eta)$$

$$\psi_4(\boldsymbol{\zeta}) = \psi_4(\zeta,\eta) = \frac{1}{4}(1-\zeta)(1+\eta)$$

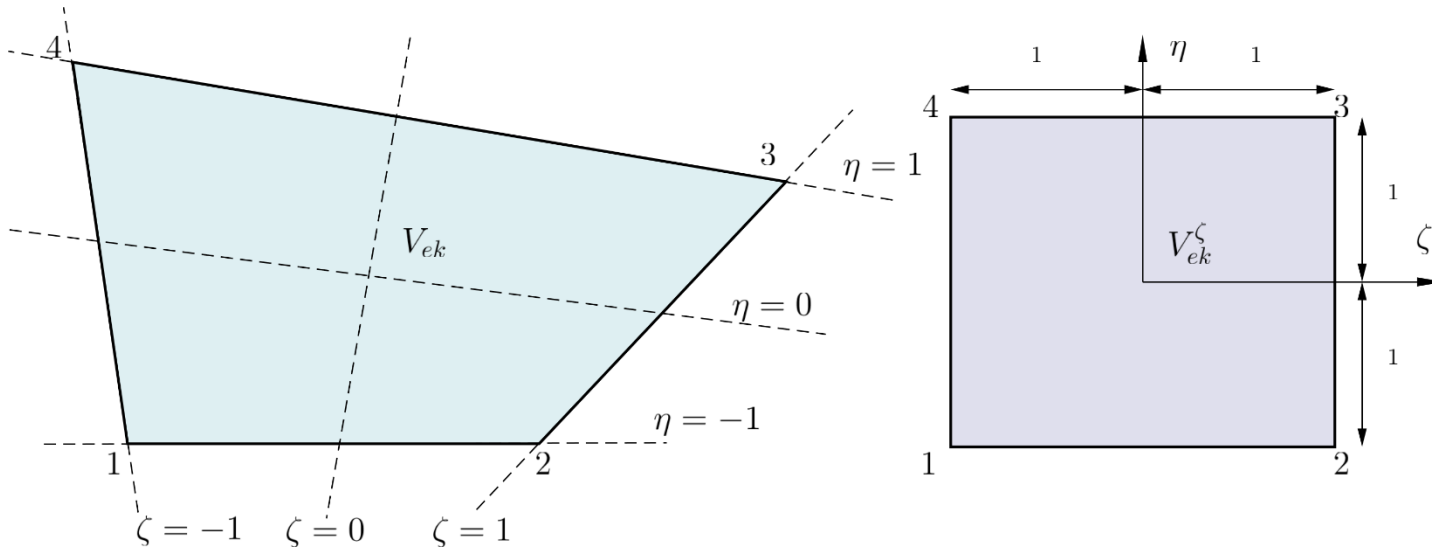


- You can see that the shape function of node i is zero at any node but i
- You can check and see that the shape function of node i is 1 only at node i

Finite Element Method – Iso-parametric elements

- The idea is to convert the integrals over arbitrary-shaped elements (left) to an integral over natural elements (right), because computing the integrals over the natural domain is easy. Remember how we can convert the domain of integration:

$$\int f(u)du = \int f(g(v))\frac{dg}{dv}dv$$



Finite Element Method – Iso-parametric elements

- Substitute $v = \zeta$, $g = \boldsymbol{x}$ and $u = \boldsymbol{x}(\zeta)$ to get:

$$\int_{V_e} f(\boldsymbol{x}(\boldsymbol{\zeta}))d\boldsymbol{x} = \int_{V_e^n} f(\boldsymbol{x}(\boldsymbol{\zeta})))|\frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}}|d\boldsymbol{\zeta}$$

- Here, $\boldsymbol{J} = \frac{dx}{d\zeta} = \frac{\partial x}{\partial \zeta} = \frac{\partial x(x,y,z)}{\partial \zeta(\zeta,\eta,\chi)}$ is the Jacobian matrix of the transformation
- It arises from the transformation and is related to the chain rule as

$$\frac{\partial}{\partial \zeta} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \zeta} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \zeta} + \frac{\partial}{\partial z}\frac{\partial z}{\partial \zeta}$$

$$\frac{\partial}{\partial \eta} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \eta} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \eta} + \frac{\partial}{\partial z}\frac{\partial z}{\partial \eta}$$

$$\frac{\partial}{\partial \chi} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \chi} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \chi} + \frac{\partial}{\partial z}\frac{\partial z}{\partial \chi}$$

Finite Element Method – Iso-parametric elements

- In the matrix form it can be stated as

$$\frac{d}{d\boldsymbol{\zeta}} = \frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}} \frac{d}{d\boldsymbol{x}}$$

$$\begin{bmatrix} \frac{\partial}{\partial \zeta} \\ \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial \chi} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \chi} & \frac{\partial y}{\partial \chi} & \frac{\partial z}{\partial \chi} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \boldsymbol{J} \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}$$

- In 2D as $\boldsymbol{J} = \begin{bmatrix} \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix}$
- And in 1D as $\boldsymbol{J} = \begin{bmatrix} \frac{\partial x}{\partial \zeta} \end{bmatrix}$

Finite Element Method – Iso-parametric elements

- Conversely, we can write $\frac{d}{d\boldsymbol{x}}$ as

$$\frac{d}{d\boldsymbol{x}} = \left(\frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}}\right)^{-1} \frac{d}{d\boldsymbol{\zeta}}$$

$$\frac{d}{d\boldsymbol{x}} = \boldsymbol{J}^{-1} \frac{d}{d\boldsymbol{\zeta}}$$

- Why should you care? Because this relationship is needed to convert the higher-order space-derivatives from the element's physical coordinates to natural coordinates

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \frac{d}{d\boldsymbol{x}} = \boldsymbol{J}^{-1}\frac{d}{d\boldsymbol{\zeta}} = \boldsymbol{J}^{-1}\nabla_{\boldsymbol{\zeta}} = \boldsymbol{J}^{-1} \begin{bmatrix} \frac{\partial}{\partial \zeta} \\ \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial \chi} \end{bmatrix}$$

Finite Element Method – Iso-parametric elements

- Note that the Jacobian matrix is evaluated for each element as follows:

$$\boldsymbol{J} = \boldsymbol{J}(\boldsymbol{\zeta}) = \boldsymbol{J}(\zeta, \eta, \chi) = \begin{bmatrix} \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \chi} & \frac{\partial y}{\partial \chi} & \frac{\partial z}{\partial \chi} \end{bmatrix}$$

$$= \begin{bmatrix} \sum_j x_j(t) \frac{\partial \psi_j}{\partial \zeta} & \sum_j y_j(t) \frac{\partial \psi_j}{\partial \zeta} & \sum_j z_j(t) \frac{\partial \psi_j}{\partial \zeta} \\ \sum_j x_j(t) \frac{\partial \psi_j}{\partial \eta} & \sum_j y_j(t) \frac{\partial \psi_j}{\partial \eta} & \sum_j z_j(t) \frac{\partial \psi_j}{\partial \eta} \\ \sum_j x_j(t) \frac{\partial \psi_j}{\partial \chi} & \sum_j y_j(t) \frac{\partial \psi_j}{\partial \chi} & \sum_j z_j(t) \frac{\partial \psi_j}{\partial \chi} \end{bmatrix}$$

Finite Element Method – Iso-parametric elements

- Finally, the iso-parametric form of the element matrices can be calculated as follows:
- Mass matrix elements

$$\int_{V_{e_k}} \psi_1(\boldsymbol{x})\psi_1(\boldsymbol{x})d\boldsymbol{x} = \int_{V_{e_k}^{\boldsymbol{\zeta}}} \psi_1(\boldsymbol{x}(\boldsymbol{\zeta}))\psi_2(\boldsymbol{x}(\boldsymbol{\zeta}))|\frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}}|d\boldsymbol{\zeta}$$

$$= \int_{\eta=-1}^{\eta=1}\int_{\zeta=-1}^{\zeta=1} \psi_1(\boldsymbol{\zeta})\psi_2(\boldsymbol{\zeta})|\boldsymbol{J}|\ d\zeta d\eta$$

- Note that the whole procedure was necessary to transform to the integrals from $\int_y^y \int_x^x$ to $\int_{\eta=-1}^{\eta=+1} \int_{\zeta=-1}^{\zeta=+1}$, so that it can be numerically computed

Finite Element Method – Iso-parametric elements

- Elements of the advection matrix can be computed via

$$\int_{V_{e_k}} \psi_1(\boldsymbol{x}) \nabla \psi_1(\boldsymbol{x}) d\boldsymbol{x} = \int_{V_{e_k}^{\boldsymbol{\zeta}}} \psi_1(\boldsymbol{x}(\boldsymbol{\zeta})) \Big(\frac{d\boldsymbol{\zeta}}{d\boldsymbol{x}} \nabla_{\boldsymbol{\zeta}} \psi_1 \big(\boldsymbol{x}(\boldsymbol{\zeta})\big) \Big) |\frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}}| d\boldsymbol{\zeta}$$

$$= \int_{\eta=-1}^{\eta=1} \int_{\zeta=-1}^{\zeta=1} \psi_1(\boldsymbol{\zeta}) \Big(\boldsymbol{J}^{-1} \nabla_{\boldsymbol{\zeta}} \psi_1(\boldsymbol{\zeta}) \Big) |\boldsymbol{J}| \; d\zeta d\eta$$

- Elements of the diffusion matrix can be computed via

$$\int_{V_{e_k}} \nabla \psi_1(\boldsymbol{x}) \cdot (\boldsymbol{D} \nabla \psi_1(\boldsymbol{x})) d\boldsymbol{x} = \int_{V_{e_k}^{\boldsymbol{\zeta}}} \left(\Big(\frac{d\boldsymbol{\zeta}}{d\boldsymbol{x}} \nabla_{\boldsymbol{\zeta}} \psi_1 \big(\boldsymbol{x}(\boldsymbol{\zeta})\big) \Big) \cdot \boldsymbol{D} \Big(\frac{d\boldsymbol{\zeta}}{d\boldsymbol{x}} \nabla_{\boldsymbol{\zeta}} \psi_1 \big(\boldsymbol{x}(\boldsymbol{\zeta})\big) \Big) \right) |\frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}}| d\boldsymbol{\zeta}$$

$$= \int_{\eta=-1}^{\eta=1} \int_{\zeta=-1}^{\zeta=1} \left(\Big(\boldsymbol{J}^{-1} \nabla_{\boldsymbol{\zeta}} \psi_1(\boldsymbol{\zeta}) \Big) \cdot \boldsymbol{D} \Big(\boldsymbol{J}^{-1} \nabla_{\boldsymbol{\zeta}} \psi_1(\boldsymbol{\zeta}) \Big) \right) |\boldsymbol{J}| \; d\zeta d\eta$$

Finite Element Method – Iso-parametric elements

- Elements of the load vector can be computed via

$$\int_{V_{e_k}} \psi_1(\boldsymbol{x}) s(\boldsymbol{x}) d\boldsymbol{x} = \int_{V_{e_k}^{\boldsymbol{\zeta}}} \psi_1\big(\boldsymbol{x}(\boldsymbol{\zeta})\big) s\big(\boldsymbol{x}(\boldsymbol{\zeta})\big) \left|\frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}}\right| d\boldsymbol{\zeta} = \int_{\eta=-1}^{\eta=1} \int_{\zeta=-1}^{\zeta=1} \psi_1(\boldsymbol{\zeta}) s(\boldsymbol{\zeta}) |\boldsymbol{J}| \, d\zeta d\eta$$

- Similarly, the surface integral (line integral in 2D) that comes into play if the element has a segment on the boundary

$$\int_{S_N \cap S_{e_k}} \psi_1(\boldsymbol{x}) h d\boldsymbol{x} = \int_{S_N \cap S_{e_k}^{\boldsymbol{\zeta}}} \psi_1\big(\boldsymbol{x}(\boldsymbol{\zeta})\big) h \left|\frac{d\boldsymbol{x}}{d\boldsymbol{\zeta}}\right| d\boldsymbol{\zeta} = \int_{\zeta=-1}^{\zeta=1} \psi_1(\boldsymbol{\zeta}) h |\boldsymbol{J}| \, d\zeta$$

- $h = (\boldsymbol{u}\bar{\phi} - D \ \nabla\bar{\phi}) \cdot \boldsymbol{n}$ is the boundary flux on the Neumann part of the boundary

Finite Element Method – Gauss-Quadrature

- The integrals were converted to the natural domain, but how do we actually compute them?
- Gauss-Quadrature is a numerical method for approximating an integral as follows:

$$\int_{x=-1}^{x=+1} f(x)dx = \sum_{i=1}^{n} w_i f(x_i)$$

- The idea is to approximate the integral according to the value of the integrand $f(x)$ and a weight function w at "some" integration points i
- n quadrature points can exactly compute the integral for a polynomial function f of degree $2n-1$ or less, e.g., 3 points approximation can exactly compute the integral of a polynomial of degree 5.

Finite Element Method – Gauss-Quadrature

- Examples

n	x_i	w_i
1	0	2
2	$-\frac{1}{\sqrt{3}}, +\frac{1}{\sqrt{3}}$	-0.57735,-0.57735
3	$-\frac{3}{\sqrt{5}}, 0, +\frac{3}{\sqrt{5}}$	$\frac{5}{9}, \frac{8}{9}, \frac{5}{9}$

- The same method can be used for 2D or 3D integrals as follows:

$$\int_{y=-1}^{y=+1}\int_{x=-1}^{x=+1} f(x,y)dxdy = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j f(x_i, y_j)$$

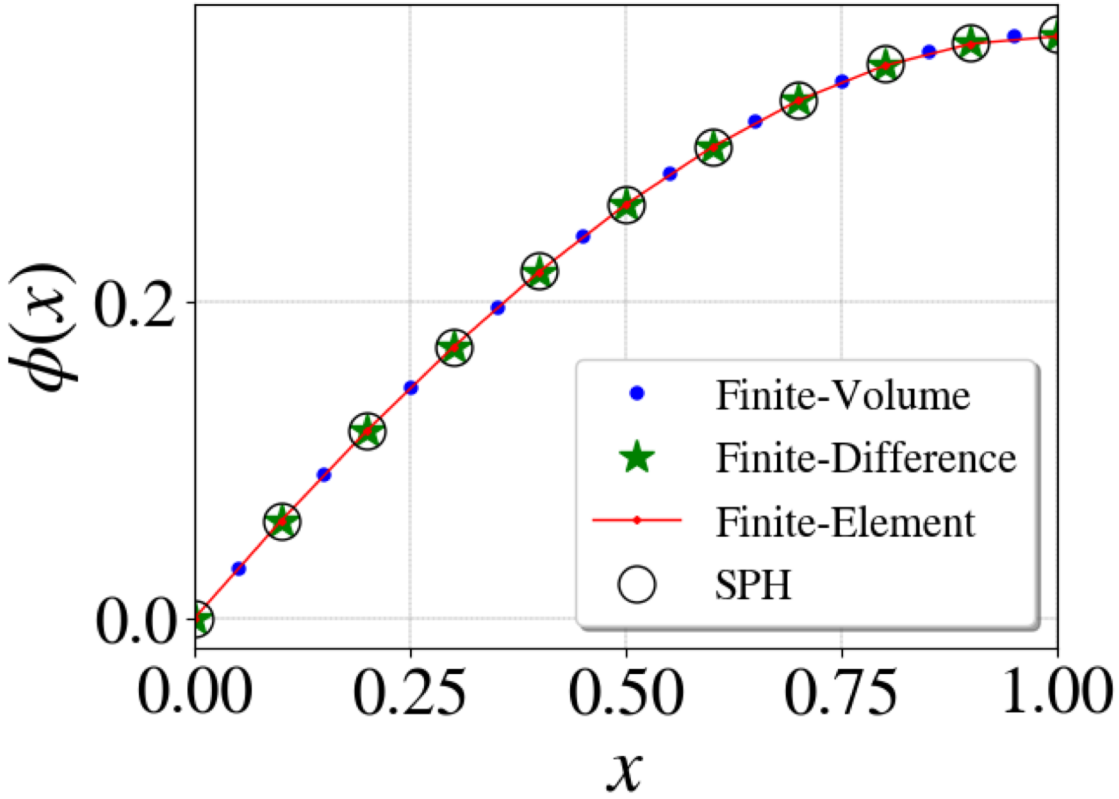
$$\int_{z=-1}^{z=+1}\int_{y=-1}^{y=+1}\int_{x=-1}^{x=+1} f(x,y)dxdydz = \sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n} w_i w_j w_k f(x_i, y_j, z_k)$$

Space Discretization– Final Thoughts

- FEM calculations are more involved than FD, FV or SPH methods, yet they offer an easy way to increase the order of accuracy in space
- This is possible if one uses higher-order elements, all the calculations are similar and limited to each element
- This is not the case in FD and FV, because obtaining a high-order accurate solution requires extending the interpolations beyond the first-neighboring elements
- Ultimately, you should view all of these methods as space-discretization techniques, which aim at approximating the space-derivative terms in a PDE

Space Discretization– Final Thoughts

- Each discretization method comes with advantages and disadvantages
- Choosing the proper method depends on the application of interest
- Exercise: solve the 1D steady conservation equation via these methods



Computational Fluid Dynamics

- Note that mass and momentum balance describe the time evolution of any continua
- However, they should be solved for every point in the domain
- To make it possible, we chose to solve these equations in **discrete points** in space
- Computational Fluid Dynamics (CFD) is the study of methods for solving these equations with appropriate **space discretization** and **time integration** schemes
- Since this is just an introduction, we explain a few discretization techniques only for the general conservation law, yet they are applicable to Navier-Stokes equations

Conservation Principles – Summary

- If there are two equations that you need to know by heart, they are
  1. The conservation law for any conserved variable ϕ:

$$\underbrace{\frac{\partial \phi(\boldsymbol{x},t)}{\partial t}}_{\text{rate of change of } \phi} = \underbrace{-\nabla \cdot \boldsymbol{f}(\boldsymbol{x},t)}_{\text{sink or source of the flux of } \phi} + \underbrace{s(\boldsymbol{x},t)}_{\text{rate of generation or dissipation of } \phi}$$

  2. The flux of ϕ, which often can be written as the sum of the advective and diffusive flux

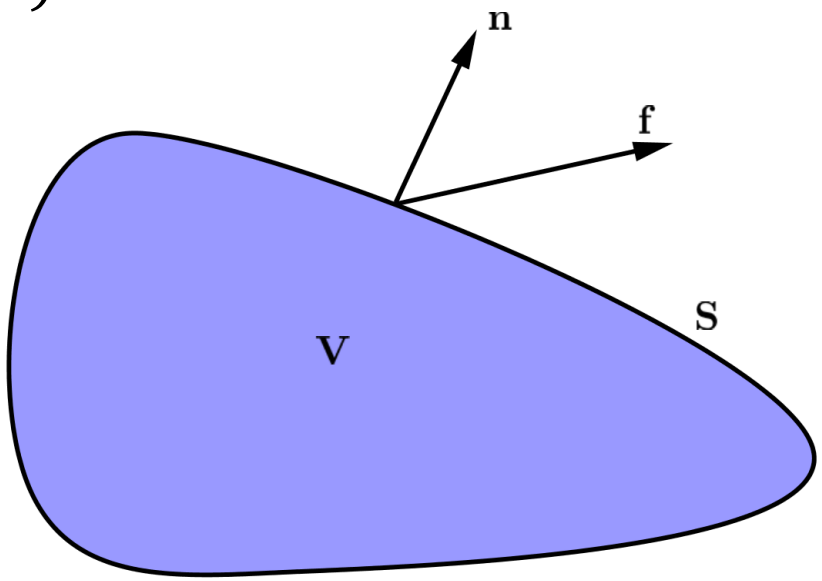
$$\boldsymbol{f}(\boldsymbol{x},t) = \boldsymbol{u}(\boldsymbol{x},t)\phi(\boldsymbol{x},t) - \boldsymbol{D}(\boldsymbol{x},t)\nabla\phi(\boldsymbol{x},t)$$
$$\boldsymbol{f} = \boldsymbol{u}\phi - \boldsymbol{D}\nabla\phi$$

Conservation Principles – Flashback

- Imagine a **control volume** ($\boldsymbol{V}$) surrounded by the **control surfaces** ($\boldsymbol{S}$)
- Denote $c = c(\boldsymbol{x}, t)$ as the concentration of a conserved scalar variable C per **unit mass** of the control volume:

$$C = \int_V c(\boldsymbol{x}, t) dm = \int_V c(\boldsymbol{x}, t)\rho(\boldsymbol{x}, t) d\boldsymbol{x} = \int_V \phi(\boldsymbol{x}, t) d\boldsymbol{x}$$

- Define $\phi(\boldsymbol{x}, t) = \rho(\boldsymbol{x}, t)c(\boldsymbol{x}, t)$ to be the **volume** concentration of C, i.e. C per **unit volume**



Conservation Principles – Mass Conservation

- A special case of the conservation principle when the conserved property is mass

1. In this case C= m, i.e., c=dm/dm=1, and $\phi = \rho c = \rho$

2. Mass is transported only by the velocity field, and not molecular diffusion contribute to mass transport, so the flex of mass is only advective, i.e., $\boldsymbol{f} = \boldsymbol{u}\phi = \boldsymbol{u}\rho$

3. No mass source or sink, so $s(\boldsymbol{x}, t) = 0$

Conservation Principles – Mass Conservation

- Use the previous information in the conservation law for any conserved variable ϕ:

$$\underbrace{\frac{\partial \phi(\boldsymbol{x},t)}{\partial t}}_{\text{rate of change of } \phi} = \underbrace{-\nabla \cdot \boldsymbol{f}(\boldsymbol{x},t)}_{\text{sink or source of the flux of } \phi} + \underbrace{s(\boldsymbol{x},t)}_{\text{rate of generation or dissipation of } \phi}$$

- Leads to different forms of mass conservation

$$\text{general form: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0 \tag{1}$$

$$\text{Eulerian form: } \frac{\partial \rho}{\partial t} + \boldsymbol{u} \cdot \nabla \rho + \rho \nabla \cdot \boldsymbol{u} = 0 \tag{2}$$

$$\text{Lagrangian form: } \frac{D\rho}{Dt} + \rho \nabla \cdot \boldsymbol{u} = 0 \tag{3}$$

Mass Conservation – Incompressible flow

- For incompressible flow, total time-variation of density is zero $\frac{D\rho}{Dt} = 0$
- The mass conservation (e.g., equation 3), is simplified to

$$\nabla \cdot \boldsymbol{u} = 0$$

- Note that this equation is **not** mass conservation for any flows. It is **not** the incompressible flow assumption. It is mass conservation only for incompressible flows
- Remember the property of the divergence operator, zero-divergence means no source/sink
- This makes sense because if the density of a media is constant everywhere, then there could be no source/sink in the velocity field
- Note that we assumed there is no mass source or sink in the domain, e.g., we are not pumping in fluid into the domain

Conservation Principles – Momentum Conservation

- A special case of the conservation principle when the conserved property is momentum

1. In this case, C= $m\boldsymbol{u}$, i.e., $c = \boldsymbol{u}$, and $\boldsymbol{\phi} = \rho \boldsymbol{c} = \rho \boldsymbol{u}$ is a vector property

2. Note that here, the conserved property is a vector, rather than a scalar. However, we can still use the same equations. Just think about a vector as 3 scalars. We are going to derive a vector equation, or 3 scalar equations

3. Assume that there is a source/sink for creating momentum, such as $\boldsymbol{s}(\boldsymbol{x}, t) = \boldsymbol{f}_b$ Usually, this how body forces such as gravity come into play

Momentum Conservation – Fluxes

- The flux of scalar quantity was a vector
- The flux of a vector is a tensor (second-order)
- In this case, momentum is transferred via velocity and stress tensor

$$\boldsymbol{f} = \boldsymbol{u}(\boldsymbol{x},t)\phi(\boldsymbol{x},t) - \boldsymbol{\sigma}(\boldsymbol{x},t) = \rho\boldsymbol{u}\boldsymbol{u} - \boldsymbol{\sigma}$$

$$= \rho \begin{bmatrix} u_1u_1 & u_1u_2 & u_1u_3 \\ u_2u_1 & u_2u_2 & u_2u_3 \\ u_3u_1 & u_3u_2 & u_3u_3 \end{bmatrix} + \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

- $\boldsymbol{uu}$ is a tensor resulting from the dyadic product of $\boldsymbol{u}$ by itself, and $\boldsymbol{\sigma}$ is the stress flux

- This is not only applicable to fluids, but any other continuous media

Momentum Conservation – Stress Decomposition

- For the sake of modeling, we decompose stress flux into normal stress and shear stress

- These components are
  - Normal stress or **volumetric**/hydrostatic part, it is diagonal tensor formed from the trace of stress
  - Shear stress or **deviatoric** part, a traceless tensor

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{dev} + \boldsymbol{\sigma}_{vol}$$
$$tr(\boldsymbol{\sigma}_{vol}) = tr(\boldsymbol{\sigma}) = \sigma_{11} + \sigma_{22} + \sigma_{33}$$
$$tr(\boldsymbol{\sigma}_{dev}) = 0$$

- $tr(\quad) = (\quad) : I$ indicates the sum of diagonal elements of a tensor

Momentum Conservation – Stress Decomposition

- Define pressure as the average of the trace of the stress

$$p = -(\sigma_{11} + \sigma_{22} + \sigma_{33})/3$$

- Simplify the decomposition via:

$$\boldsymbol{\sigma}_{vol} = (\frac{\boldsymbol{\sigma} : \mathbf{I}}{3})\mathbf{I} = (\frac{\sigma_{11} + \sigma_{22} + \sigma_{33}}{3})\mathbf{I} = -p \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{\sigma}_{dev} = \boldsymbol{\sigma} - (\frac{\boldsymbol{\sigma} : \mathbf{I}}{3})\mathbf{I} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} + \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix}$$

Momentum Conservation

- Substitute in the flux, $\boldsymbol{\phi} = \rho \boldsymbol{u}$ and $\boldsymbol{s} = \boldsymbol{f}_b$ into the conservation law

$$\underbrace{\frac{\partial \phi(\boldsymbol{x}, t)}{\partial t}}_{\text{rate of change of } \phi} = \underbrace{-\nabla \cdot \boldsymbol{f}(\boldsymbol{x}, t)}_{\text{sink or source of the flux of } \phi} + \underbrace{s(\boldsymbol{x}, t)}_{\text{rate of generation or dissipation of } \phi}$$

- The general form of momentum balance for any continuum

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}\boldsymbol{u} - \boldsymbol{\sigma}_{vol} - \boldsymbol{\sigma}_{dev}) - \boldsymbol{f}_b = 0$$

Momentum Conservation- Simplifications

1. For incompressible flows remember $\nabla \cdot \boldsymbol{u} = 0$, so the advective term is simplified to

$$\nabla \cdot (\rho \boldsymbol{u}\boldsymbol{u}) = \boldsymbol{u} \cdot \nabla(\rho \boldsymbol{u}) + \rho \boldsymbol{u}(\nabla \cdot \boldsymbol{u}) = \boldsymbol{u} \cdot \nabla(\rho \boldsymbol{u})$$

2. Remember the form of divergence of a tensor. Divergence of the volumetric stress tensor is simplified to

$$\nabla \cdot (\boldsymbol{\sigma}_{vol}) = \nabla \cdot (p\mathbf{I}) = \nabla p$$

- The final form of the momentum balance for any continuum becomes a **vector** equation. You can think about it as 3 scalar equations for each component of $\boldsymbol{u}$

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla(\rho \boldsymbol{u}) + \nabla p + \nabla \cdot (\boldsymbol{\sigma}_{dev}) - \boldsymbol{f}_b = 0$$

Mass & Momentum Conservation – Final Thoughts

1. Note that with mass and momentum balance we have 4 equations

$$\text{One equation: } \frac{\partial \rho}{\partial t} + \boldsymbol{u} \cdot \nabla \rho + \rho \nabla \cdot \boldsymbol{u} = 0$$

$$\text{Three equations: } \frac{\partial \rho \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla (\rho \boldsymbol{u}) + \nabla p + \nabla \cdot (\boldsymbol{\sigma}_{dev}) - \boldsymbol{f}_b = 0$$

2. We have 13 unknowns (3 for $\boldsymbol{u}$, p, and 9 components of $\boldsymbol{\sigma}_{dev}$)
3. We need to find additional equations for closure
4. The idea is to use information about the **material** to model $\boldsymbol{\sigma}_{dev}$
5. From here we are parting with general continua and seek additional equations when working with Newtonian fluids

Material Model- Newtonian Fluids

1. In Newtonian fluids, the deviatoric stress tensor is a linear function of deformation gradient, i.e., $\boldsymbol{\sigma}_{dev} = 2\mu\boldsymbol{E}$

2. Deformation gradient is obtained from velocity gradient tensor via

$$\boldsymbol{E} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^{\mathbf{T}})$$

3. Remember the form of the gradient for a vector,

$$\nabla \boldsymbol{u} = \nabla \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} & \frac{\partial u_x}{\partial z} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} & \frac{\partial u_y}{\partial z} \\ \frac{\partial u_z}{\partial x} & \frac{\partial u_z}{\partial y} & \frac{\partial u_z}{\partial z} \end{bmatrix}$$

Material Model- Newtonian Fluids

- Remember the form of divergence for tensors
- Simplify $\nabla \cdot (\boldsymbol{\sigma}_{dev}) = \nabla \cdot \ (\mu \ (\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T))$

$$\nabla \cdot (\nabla \boldsymbol{u}) = \begin{bmatrix} \frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial y^2} + \frac{\partial^2 u_x}{\partial z^2} \\ \frac{\partial^2 u_y}{\partial x^2} + \frac{\partial^2 u_y}{\partial y^2} + \frac{\partial^2 u_y}{\partial z^2} \\ \frac{\partial^2 u_z}{\partial x^2} + \frac{\partial^2 u_z}{\partial y^2} + \frac{\partial^2 u_z}{\partial z^2} \end{bmatrix} = \nabla^2(\boldsymbol{u})$$

$$\nabla \cdot (\nabla \boldsymbol{u}^T) = \begin{bmatrix} \frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_y}{\partial x \partial y} + \frac{\partial^2 u_z}{\partial x \partial z} \\ \frac{\partial^2 u_x}{\partial x \partial y} + \frac{\partial^2 u_y}{\partial y^2} + \frac{\partial^2 u_z}{\partial y \partial z} \\ \frac{\partial^2 u_x}{\partial x \partial z} + \frac{\partial^2 u_z}{\partial y \partial z} + \frac{\partial^2 u_z}{\partial z^2} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x}(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}) \\ \frac{\partial}{\partial y}(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}) \\ \frac{\partial}{\partial z}(\frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial y} + \frac{\partial u_z}{\partial z}) \end{bmatrix} = \nabla(\nabla \cdot \boldsymbol{u}) = \mathbf{0}$$

Momentum Conservation - Newtonian Incompressible Fluids

- Note that Laplacian is $\nabla^2(\quad) = \frac{\partial^2(\quad)}{\partial x^2} + \frac{\partial^2(\quad)}{\partial y^2} + \frac{\partial^2(\quad)}{\partial z^2}$

- We related 9 unknown in deviatoric stress tensor to three components of the velocity and can write

$$\nabla.(\boldsymbol{\sigma}_{dev}) = \nabla.(2\mu\boldsymbol{E}) = \nabla.(\mu(\nabla\mathbf{u} + \nabla\mathbf{u}^{\mathbf{T}})) = \mu\nabla^{\mathbf{2}}\mathbf{u}$$

- The final form of the momentum equations for **incompressible Newtonian** flow (also called Navier-Stokes equations) :

$$\text{Eulerian form: } \frac{\partial\rho\boldsymbol{u}}{\partial t} + \boldsymbol{u}\bullet\nabla(\rho\boldsymbol{u}) + \nabla p - \mu\nabla^2\boldsymbol{u} - \boldsymbol{f}_b = 0$$

$$\text{Lagrangian form: } \frac{D(\rho\boldsymbol{u})}{Dt} + \nabla p - \mu\nabla^2\boldsymbol{u} - \boldsymbol{f}_b = 0$$

Numerical Solution of Fluid Flows- Lagrangian Form

- We need to solve the mass and momentum balance equations

$$\frac{D\rho}{Dt} + \rho\nabla \cdot \boldsymbol{u} = 0 \tag{1a}$$

$$\frac{D\boldsymbol{u}}{Dt} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{f}_b \tag{1b}$$

- Note that 1b is a vector equation and consists of 3 equations
- Overall, there are 4 equations and 4 unknown, i.e., $\boldsymbol{u}(\boldsymbol{x}, t) = (u_x, u_y, u_z)$ and $p = p(\boldsymbol{x}, t)$
- Note that the Lagrangian form above is linear in terms of $\boldsymbol{u}$, and p
- We just need to find out a proper time-integration technique

Numerical Solution of Fluid Flows- Eulerian Form

- The Eulerian form of the equations is as follows:

$$\nabla \cdot \boldsymbol{u} = 0 \tag{1a}$$

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla(\rho \boldsymbol{u}) = -\nabla p + \mu \nabla^2 \boldsymbol{u} + \boldsymbol{f}_b \tag{1b}$$

- Note that the Eulerian form above is non-linear in terms of $\boldsymbol{u}$, as is clear from the advection term $\boldsymbol{u} \cdot \nabla(\rho \boldsymbol{u})$
- This creates additional complications when solving for $\boldsymbol{u}$
- A typical approach to deal with this non-linearity is to use a previous-step velocity in one of the $\boldsymbol{u}$ terms in the advection term

Fluid Flow Problems- High level overview

- Two important aspects of the solution procedure are space-discretization and time-integration

- In principle time-integration of the equations and space-discretization are independent, i.e., you may use any of the space-discretization discussed before with any of time-integration that will be explained
- We will cover the simplest and well-known time-integration methods to introduce possible numerical procedures. By no means, these are the best approach
- We will cover the time-integration for Lagrangian methods, so the term $\boldsymbol{u} \cdot \nabla(\rho \boldsymbol{u})$ is not going to appear in the equations

Fluid Flow Problems- Time-Integration

- Possible integration schemes are explicit, semi-implicit, and implicit integration methods
- explicit schemes: very fast per time step, very small time-steps is needed for stability
- Implicit schemes: solve the system of equations each time step, guarantee large time-steps
- Semi-implicit schemes: a combination of explicit and implicit, usually explicit in velocity and implicit in pressure, or implicit in velocity and explicit in pressure. This usually depends on the problem of interest, e.g., highly viscous flows need implicit velocity-integration

Fluid Flow Problems- Semi-Implicit scheme

- Possible choices of integration are variant of the SIMPLE method
- SIMPLE : Semi-Implicit Method for Pressure Linked Equations

- The idea is to use the projection method and Helmholtz-Hodge decomposition

- Hodge decomposition implies that a vector field $\boldsymbol{u}(\boldsymbol{x})$ may be decomposed into a solenoidal $\boldsymbol{u}_{sol}$ and irrotational $\boldsymbol{u}_{irror}$ fields, i.e., $\boldsymbol{u} = \boldsymbol{u}_{sol} + \boldsymbol{u}_{irrot}$
- By definitions, $\nabla \cdot \boldsymbol{u}_{sol} = 0$ and $\nabla \times \boldsymbol{u}_{irrot} = 0$
- The curl-free part ($\boldsymbol{u}_{irrot}$) can be written as gradient of a scalar potential field i.e., $\boldsymbol{u}_{irrot} = \nabla\phi$
- $\boldsymbol{u}_{irrot}$ is a conservative vector field, and its line integral is path independent

Fluid Flow Problems- Hodge Decomposition

- Hodge decomposition implies that a vector field $\boldsymbol{u}(\boldsymbol{x})$ may be decomposed into a solenoidal $\boldsymbol{u}_{sol}$ and irrotational $\boldsymbol{u}_{irror}$ fields, i.e., $\boldsymbol{u} = \boldsymbol{u}_{sol} + \boldsymbol{u}_{irrot}$
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- $\boldsymbol{u}_{irrot}$ is a conservative vector field, and its line integral is path independent

- $\nabla \cdot \boldsymbol{u} = \nabla \cdot (\boldsymbol{u}_{sol} + \boldsymbol{u}_{irrot}) = \nabla \cdot \boldsymbol{u}_{irrot} = \nabla \cdot \nabla \phi = \nabla^2 \phi$
- If $\boldsymbol{u}$ is known, the Poisson equation $\nabla^2 \phi = \nabla \cdot \boldsymbol{u}$ gives ϕ, and $\boldsymbol{u}_{\text{irrot}} = \nabla \phi$ gives $\boldsymbol{u}_{irrot}$
- $\boldsymbol{u}_{sol} = \boldsymbol{u} - \boldsymbol{u}_{irrot} = \boldsymbol{u} - \nabla \phi$ gives the $\boldsymbol{u}_{sol}$

Fluid Flow Problems- Hodge Decomposition/ Navier-Stokes

- Remember the Navier-Stokes equations:

$$\frac{D\boldsymbol{u}}{Dt} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{f}_b$$

- The LHS and RHS of the Navier-Stokes are vector fields, so we can apply the Hodge decomposition
- Here, $\nu\nabla^2\boldsymbol{u} + \boldsymbol{f}_b$ is the solenoidal part, and $\frac{1}{\rho}\nabla p$ is the irrotational part, remember that the irrotational part could be written as the gradient of a scalar field. p acts similar to what we called ϕ in the previous slide

Fluid Flow Problems- Hodge Decomposition/ Navier-Stokes

- We can assume that the LHS can also be decomposed into the solenoidal and irrotational parts
- This is possible by introducing an intermediate velocity field such as $\boldsymbol{u}^*$

$$\frac{(\boldsymbol{u}^{n+1} - \boldsymbol{u}^n)}{\Delta t} = \underbrace{\frac{(\boldsymbol{u}^{n+1} - \boldsymbol{u}^*)}{\Delta t}}_{\text{solenoidal contribution}} + \underbrace{\frac{(\boldsymbol{u}^* - \boldsymbol{u})}{\Delta t}}_{\text{irrotational contribution}}$$

$$= \underbrace{-\frac{1}{\rho}\nabla p}_{\text{solenoidal contribution}} + \underbrace{\nu\nabla^2\boldsymbol{u}^n + \boldsymbol{g}}_{\text{irrotational contribution}}$$

- Note that these equations are semi-discrete; they only show time-discretization, not space-discretization

Fluid Flow Problems-Projection Method

- The two –step semi-implicit method is stated as:

$$\text{predictor} \quad \frac{(\boldsymbol{u}^* - \boldsymbol{u}^n)}{\Delta t} = +\nu\nabla^2\boldsymbol{u}^n + \boldsymbol{g} = \boldsymbol{f}^n \quad x \in \Omega, \qquad \boldsymbol{u}^* = 0 \quad x \in \partial\Omega$$

$$\text{corrector} \quad \frac{(\boldsymbol{u}^{n+1} - \boldsymbol{u}^*)}{\Delta t} = -\frac{1}{\rho}\nabla p^{n+1} \quad x \in \Omega, \qquad \nabla.\boldsymbol{u}^{n+1} = 0 \quad \in \partial\Omega$$

- In this variant velocity predictor computed explicitly, i.e., you find $\boldsymbol{u}^*$ directly from $\boldsymbol{u}^n$

$$\boldsymbol{u}^* = \boldsymbol{f}^n\Delta t + \boldsymbol{u}^n$$

- Pressure is implicit

$$\boldsymbol{u}^{n+1} = -\Delta t\frac{1}{\rho}\nabla p^{n+1} + \boldsymbol{u}^*$$

Fluid Flow Problems-Projection Method

- The important question is how to find p^{n+1}?
- The idea is to take the divergence of the irrotational part to find a scalar pressure equation

$$\frac{\nabla.\boldsymbol{u}^{n+1} - \nabla.\boldsymbol{u}^{*}}{\Delta t} = -\frac{1}{\rho}\nabla^2 p^{n+1}$$

- Note that from the incompressible flow assumption we had $\nabla \cdot \boldsymbol{u}^{n+1} = \boldsymbol{0}$
- The pressure equation becomes:

$$\frac{1}{\rho}\nabla^2 p^{n+1} = \frac{\nabla \cdot \boldsymbol{u}^{*}}{\Delta t}, \quad \nabla p^{n+1} \cdot \boldsymbol{n}|_{\partial\Omega} = 0$$

- Remember the general conservation law for ϕ; the above equation is similar to the special case of the diffusion equation with a source term and zero Neumann boundary conditions
- This pressure equation leads to solving a linear system

Projection Method – Implicit Scheme

- The implicit time-integration via projection method is similar to what we discussed

- We just need to treat the velocity prediction equation implicitly

$$\text{predictor} \quad \frac{(\boldsymbol{u}^* - \boldsymbol{u}^n)}{\Delta t} = +\nu\nabla^2\boldsymbol{u}^* + \boldsymbol{g} = \boldsymbol{f}^n \quad x \in \Omega, \qquad \boldsymbol{u}^* = 0 \quad x \in \partial\Omega$$

$$\text{corrector} \quad \frac{(\boldsymbol{u}^{n+1} - \boldsymbol{u}^*)}{\Delta t} = -\frac{1}{\rho}\nabla p^{n+1} \quad x \in \Omega, \qquad \nabla.\boldsymbol{u}^{n+1} = 0 \quad \in \partial\Omega$$

- Note that $\nu\nabla^2\boldsymbol{u}^*$ is used in the RHS of the predictor equation instead of $\nu\nabla^2\boldsymbol{u}^n$
- This makes the formulation implicit-in-velocity, which is required integration scheme if the viscous term is comparable with the pressure gradient term

Projection Method – Crank-Nicolson

- We could think about semi-implicit prediction schemes by combining the explicit-in-velocity scheme and implicit-in-velocity scheme as :

$$\text{predictor} \quad \frac{(\boldsymbol{u}^* - \boldsymbol{u}^n)}{\Delta t} = +\nu(\theta\nabla^2\boldsymbol{u}^* + (1-\theta)\nabla^2\boldsymbol{u}^n) + \boldsymbol{g} = \boldsymbol{f}^n \quad x \in \Omega, \qquad \boldsymbol{u}^* = 0 \quad x \in \partial\Omega$$

$$\text{corrector} \quad \frac{(\boldsymbol{u}^{n+1} - \boldsymbol{u}^*)}{\Delta t} = -\frac{1}{\rho}\nabla p^{n+1} \quad x \in \Omega, \qquad \nabla.\boldsymbol{u}^{n+1} = 0 \quad \in \partial\Omega$$

- $\theta = 0$ leads to the explicit-in-velocity scheme
- $\theta = 1$ leads to the implicit-in-velocity scheme
- $\theta = \frac{1}{2}$, the so-called Crank-Nicolson, is half-implicit-in-velocity
- Note that $\theta > 1$ requires solving a system of equations to find u^*

Time-Integration, Final Thoughts

- Note that in principle the time-integration is independent of the space-discretization
- You may use any of the space-discretization methods discussed before to space-discretize the semi-discrete time-integration methods we just explained
- Many important details about the accuracy and the stability of these methods were not discussed
- You should take a CFD course to learn about those details

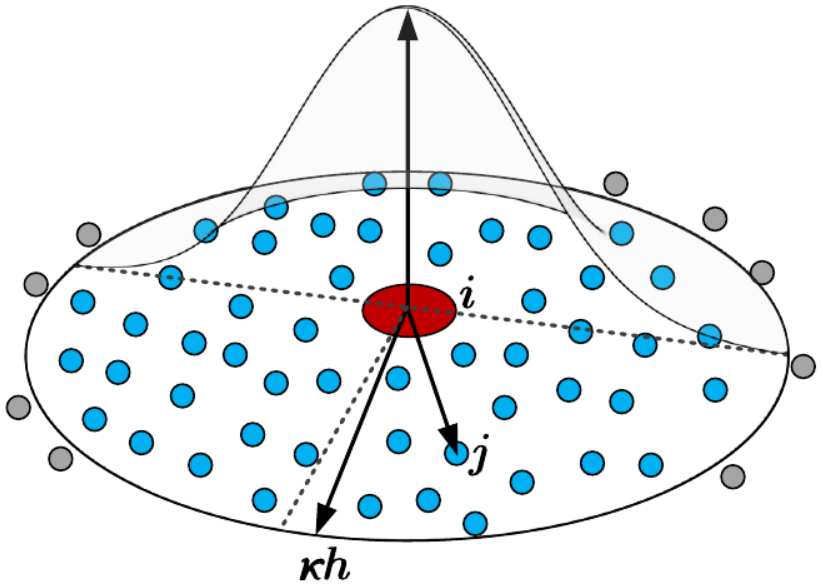
A Simple example, Weakly Compressible SPH

- Remember the SPH discretization

$$f(\mathbf{r}_i) \approx \langle f \rangle_i = \sum_{j \in \mathcal{S}(i)} \frac{m_j}{\rho_j} f(\mathbf{r}_j) W_{ij}$$

$$\nabla f(\mathbf{r}_i) = \langle \nabla f \rangle_i = \sum_{j \in \mathcal{S}(i)} V_j (f_j - f_i) \nabla_i W_{ij}$$

$$\nabla^2 f(\mathbf{r}_i) = \langle \nabla^2 f \rangle_i = 2 \sum_{j \in \mathcal{S}(i)} V_j (\mathbf{e}_{ij} \cdot \nabla_i W_{ij}) \frac{f_i - f_j}{|\mathbf{r}_{ij}|}$$



- m_j and ρ_j are the mass and density of the marker j, and $V_j = m_j/\rho_j$
- $q = r/h$ and $r = |\boldsymbol{r}_i - \boldsymbol{r}_j|$
- h is a characteristic length; it usually is the initial distance of markers

A Simple example, Weakly Compressible SPH

- W_{ij} is the weight function

$$W(|\mathbf{r}|,h)=\frac{5}{14\pi h^3}\times\begin{cases}(2-q)^3-4(1-q)^3, & 0\le q<1\\ (2-q)^3, & 1\le q<2\\ 0, & q\ge 2\end{cases}$$

- $\nabla_i W_{ij}$ is the kernel gradient

$$\nabla_i W_{ij}=\frac{\mathbf{r}_{ij}}{|\mathbf{r}_{ij}|}\frac{\partial W}{\partial q}\frac{\partial q}{\partial|\mathbf{r}_{ij}|}\bigg|_{i,j}=\frac{-15\mathbf{r}_{ij}}{14\pi h^5 q}\times\begin{cases}(2-q)^2-4(1-q)^2, & 0\le q<1\\ (2-q)^2, & 1\le q<2\\ 0, & q\ge 2\end{cases}$$

A Simple example, Weakly Compressible SPH

- The density and the volume of each marker can be computed from

$$\rho_i = \sum_{j \in \mathcal{S}(i)} m_j W_{ij} \qquad V_i = \left(\sum_{j \in \mathcal{S}(i)} W_{ij} \right)^{-1}$$

- Weakly compressible SPH go around solving the pressure equation by using an equation of state to find the pressure of marker i, i.e.,

$$p_i = (k|\mathbf{u}|_{max})^2 \left(\frac{\rho_i}{\rho_0} - 1 \right) + p_0$$

- Larger $|\boldsymbol{u}_{max}|$ is the maximum velocity in the domain, and ρ_0 and p_0 are reference density and pressure
- k is a constant represents the stiffness of the equation
- This is convenient because the pressure is computed explicitly from the density

Weakly Compressible SPH

- The equation of state allows for controlling the compressibility of the fluid through the constant k
- To have incompressible fluid, large k values should be used
- The larger the k, the stiffer the equation of state
- Large values of k demand smaller time steps; i.e., in order to get the ideal incompressible behavior we are forced to use small time steps

Weakly Compressible SPH

- Navier-Stokes equations

$$\frac{D\boldsymbol{u}}{Dt} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{f}_b,$$

- Given that the pressure is calculated explicitly, the semi-discrete form of Navier-Stokes is

$$\frac{d\mathbf{v}_i}{dt} = -\sum_{j\in\mathcal{S}(i)} \frac{1}{\rho_j^n}(p_j^n - p_i^n)\nabla_i W_{ij} V_j + 2\sum_{j\in\mathcal{S}(i)} V_j(\mathbf{e}_{ij}\cdot\nabla_i W_{ij})\frac{\mathbf{v}_i^n - \mathbf{v}_j^n}{|\mathbf{r}_{ij}|} + \mathbf{f}_b$$

- A forward-Euler integration allows for finding the new velocities as

$$\mathbf{v}_i^{n+1} = (\frac{d\mathbf{v}_i}{dt})\Delta t + \mathbf{v}_i^n$$

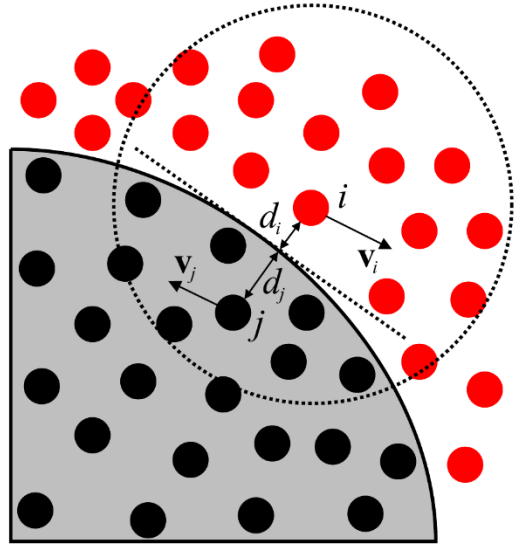
Weakly Compressible SPH- Boundary Conditions Velocity

- Impose boundary conditions via several layers of ghost/fixed particles (black markers)
- If i is a fluid particle and j a solid ghost particle, the velocity of j is computed as

$$\mathbf{v}_j = \frac{d_j}{d_i}(\mathbf{v}_j^p - \mathbf{v}_i) + \mathbf{v}_j^p$$

$$d_i = \kappa h(2\chi_i - 1) \qquad d_j = \kappa h(2\chi_j - 1)$$

$$\chi_i = \frac{\sum_{k\in\text{fluid}} W_{ik}}{\sum_k W_{ik}} \qquad \chi_j = \frac{\sum_{k\in\text{solid}} W_{jk}}{\sum_k W_{jk}}$$



- $\kappa = 2$ is the kernel compart support, and V_j^p is the prescribed velocity of the wall at the position of maker j
- Note that in this approach the velocity of a boundary maker is assigned differently for different fluid markers

Weakly Compressible SPH- Boundary Conditions

- A more convenient approach for assigning a unique value to each boundary marker *j*, regardless of the interacting fluid marker, is

$$\mathbf{v}_j = 2\mathbf{v}_j^p - \tilde{\mathbf{v}}_j$$

$$\tilde{\mathbf{v}}_j = \frac{\sum_{i \in \mathbf{F}} \mathbf{v}_i W_{ij}}{\sum_{i \in \mathbf{F}} W_{ij}}$$

$$p_j = \frac{\sum_{i \in \mathbf{F}} p_i W_{ij} + \left(\mathbf{g} - \mathbf{a}_j^p\right) \cdot \sum_{i \in \mathbf{F}} \rho_i \mathbf{r}_{ij} W_{ij}}{\sum_{i \in \mathbf{F}} W_{ij}}$$

Kinematically Constrained SPH

- Remember the earlier discussion in the course about the Newton-Euler equations of motion

$$
\begin{aligned}
\dot{\mathbf{q}} &= \mathbf{L}(\mathbf{q})\mathbf{v} \\
\mathbf{M}(\mathbf{q})\dot{\mathbf{v}} &= \mathbf{f}\left(t,\mathbf{q},\mathbf{v}\right) - \mathbf{g}_{\mathbf{q}}^{\mathrm{T}}(\mathbf{q},t)\lambda \; + \sum_{i\in\mathcal{A}(\mathbf{q},\delta)} \underbrace{\left(\widehat{\gamma}_{i,n}\,\mathbf{D}_{i,n} + \widehat{\gamma}_{i,u}\,\mathbf{D}_{i,u} + \widehat{\gamma}_{i,w}\,\mathbf{D}_{i,w}\right)}_{i^{th}\,\text{frictional contact force}} \\
\mathbf{0} &= \mathbf{g}(\mathbf{q},t)
\end{aligned}
$$

$$
i \in \mathcal{A}(\mathbf{q}(t),\delta) \quad : \quad \begin{cases} 0 \le \Phi_i(\mathbf{q}) \quad \perp \quad \widehat{\gamma}_{i,n} \ge 0 \\ \left(\widehat{\gamma}_{i,u},\widehat{\gamma}_{i,w}\right) = \underset{\sqrt{(\bar{\gamma}_u^i)^2+(\bar{\gamma}_w^i)^2}\le\mu_i\widehat{\gamma}_{i,n}}{\operatorname{argmin}} \mathbf{v}^T\cdot\left(\bar{\gamma}_u^i\,\mathbf{D}_{i,u} + \bar{\gamma}_w^i\,\mathbf{D}_{i,w}\right) \end{cases}
$$

- The question is how SPH can be used not in the context of Navier-Stokes but in Newton-Euler equations of motion?

Kinematically Constrained SPH

- For handling *kinematic constraints* the $g(q,t) = 0$ equation and the associated force $g_q^T(q,t)\lambda$ come into play.
- Imagine that SPH markers are 3DOF bodies
- Note that the generalized coordinates of each marker is only translation degrees of freedom
- The idea is to apply a Kinematic Constraint on each marker to mimic incompressibility

Kinematically Constrained SPH

- This is a scalar constraint $g(q,l) = C_f(q,t) = \frac{\rho_i - \rho_0}{\rho_0} = 0$ for each marker
- We need to form the Jacobian of constraint
- Take the derivative of this equation w.r.t. time

$$\dot{C}_i^f = \frac{d}{dt}(\frac{\rho_i}{\rho_0} - 1) = \frac{1}{\rho_0}\frac{d\rho_i}{dt} \approx \sum_j \frac{m_j}{\rho_0}\frac{dW_{ij}}{dt} = \sum_j \frac{m_j}{\rho_0}\frac{dW_{ij}}{d\mathbf{x}_{ij}}.\frac{d\mathbf{x}_{ij}}{dt} = \sum_j \frac{m_j}{\rho_0}\nabla W_{ij}.(\mathbf{v}_i - \mathbf{v}_j)$$

$$= -\sum_j \frac{m_j}{\rho_0}\nabla W_{ij}.(\mathbf{v}_j - \mathbf{v}_i)$$

$$= - < \nabla.\mathbf{v} >_i = 0$$

Kinematically Constrained SPH

- $\nabla_q g(\boldsymbol{q}, t) \equiv G$ is the Jacobian of the constraint and can be formed as

$$\dot{C}_i^f = \sum_j \frac{m_j}{\rho_0} \nabla_i W_{ij} \mathbf{u}_i - \sum_j \frac{m_j}{\rho_0} \nabla_i W_{ij} \mathbf{u}_j$$

$$\Rightarrow \qquad G_{ii} = \frac{1}{\rho_0} \sum_{k \neq i} m_k \nabla W_{ik} \quad \text{and} \quad G_{ij} = -\frac{m_j}{\rho_0} \nabla_i W_{ij}$$

- Each density constraint on a marker contributes to a single row in the full Jacobian matrix , which has 3n columns.

$$\mathbf{G}_i = \Big[\cdots , \underbrace{\frac{1}{\rho_0} \textstyle\sum_{j \in \mathcal{S}(i)} m_k \nabla_i W_{ik}^T}_{i^{th} \text{ element}}, \cdots , \underbrace{-\frac{m_j}{\rho_0} \nabla_i W_{ij}^T}_{j^{th} \text{ element s.t. } j \in \mathcal{S}(i)} , \cdots \Big]_{1 \times 3n}$$

Kinematically Constrained SPH- Boundary Conditions

- The idea is to enforce no-penetration condition by letting the SPH markers interact with the domain boundary through physical contact
- Hence, the red part in the Newton-Euler equations of motion is responsible for handling the boundary condition
- Imposing no-slip boundary condition is a little bit challenging and goes beyond the scope of this course

Fluid-Solid Interaction

- So far you learned about the motion of fluids and solids
- The goal is to solve problems where the motion of solid is driven by the surrounding fluid and the state of the fluid domain is affected by the motion of the solid domain
- There are many ways to numerically accomplish this, yet we will cover very beefily only one way

Fluid-Solid Interaction- Classifications

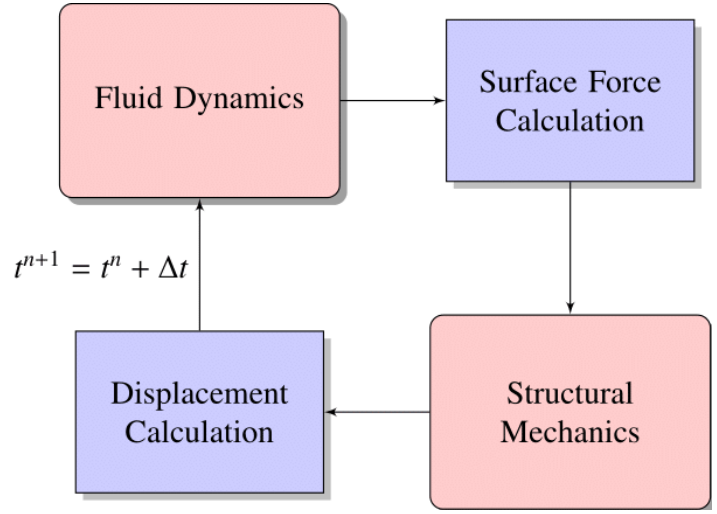
- Classification based on whether or not the sub-systems are solved simultaneously with the same formulation:

  - **Monolithic :** the fluid and the solid sub-systems are solved **simultaneously**
    - This gives you a more stable coupling and allows for larger time-steps
    - E.g., Kinematically Constrained SPH formulation; it follows the methodology of the multi-body dynamics formulation
    - Every time step, you solve for the state of fluid and rigid-bodies in one shot

  - **Partitioned :** Each time-step, you solve the solid and the fluid sub-systems **separately** and you couple the two systems together
    - It is flexible because it allows you to choose your fluid solver and the solid solver independently
    - For stability, the time-step usually is small than monolithic methods

Fluid-Solid Interaction – Partitioned methods

- The fluid solver determines the state of the ***fluid*** sub-system
- Pressure and viscose forces are calculated at the ***interface*** with the information of the updated fluid system
- The fluid forces are imposed on the solid phase as ***external forces***
- The solid sub-system is integrated as usual
- The solid sub-system updates the ***boundary*** of the fluid system



Fluid-Solid Interaction – Partitioned methods

- To see an example, consider the WCSPH method discussed earlier in the course
- Remember the velocity update formula

$$\mathbf{v}_i^{n+1} = (\frac{d\mathbf{v}_i}{dt})\Delta t + \mathbf{v}_i^n$$

- Similarly, $\mathrm{m_i}\,\frac{d\mathbf{v_i}}{\mathrm{dt}}$ for each boundary marker may be seen as the hydrodynamic force applied from the fluid to the boundary
- If one accumulates these forces on the boundary over all the boundary markers attached to the body, the total hydrodynamic force is obtained
- These forces are transferred to the multi-body solver as external forces